

Differential pair rules in gastrointestinal aGVHD

What the method resolves, what agrees with the atlas, and what is new

Each FOV-level pair is attraction (+1), avoidance (−1), or no significant rule (0). A rule is eligible only when both cell types have at least 20 cells. Net score is $100(N_{\text{attraction}} - N_{\text{avoidance}})/N_{\text{eligible}}$. Lift is observed co-occurrence divided by its independence expectation: 1 is neutral, values above 1 support attraction, and values below 1 support avoidance. **How to read the tests.** “Ordered FDR” asks whether values move consistently from Control to Mild to Severe. C–M, M–S and C–S compare the named pair of stages. “Organ FDR” compares Colon with Duodenum within one stage. FDR values are corrected for the relevant rule screen; smaller is stronger, and values below 0.05 are called significant. Temporal FDR tests the largest biopsy-level net-score gap among the three post-transplant windows; Control is shown only as context. Pooled temporal FDR jointly corrects all rules in two broader post-transplant comparisons. The discovery notebook screens every pair rule and draws a result when its best corrected test has $\text{FDR} \leq 0.05$ and its largest stage gap is at least 20 percentage points; no rule name participates in selection. The paper-alignment family screen is deliberately broader: it draws a detailed result when any corrected test has $\text{FDR} \leq 0.10$ or the Control–Severe gap is at least 30 percentage points. Rule-abundance plots use the same eligible FOVs for all three lines. These are spatial associations, not causal mechanisms.

1 Evidence that the method resolves spatial structure

The clearest separation between composition and organization is Duodenal Fibroblast → Muscle: under pathological staging the rule becomes much more avoidant while both cell populations increase, so the blue trajectory cannot be a copy of cell loss.

Evidence strength: Strong.

Duodenum Fibroblast → Muscle beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

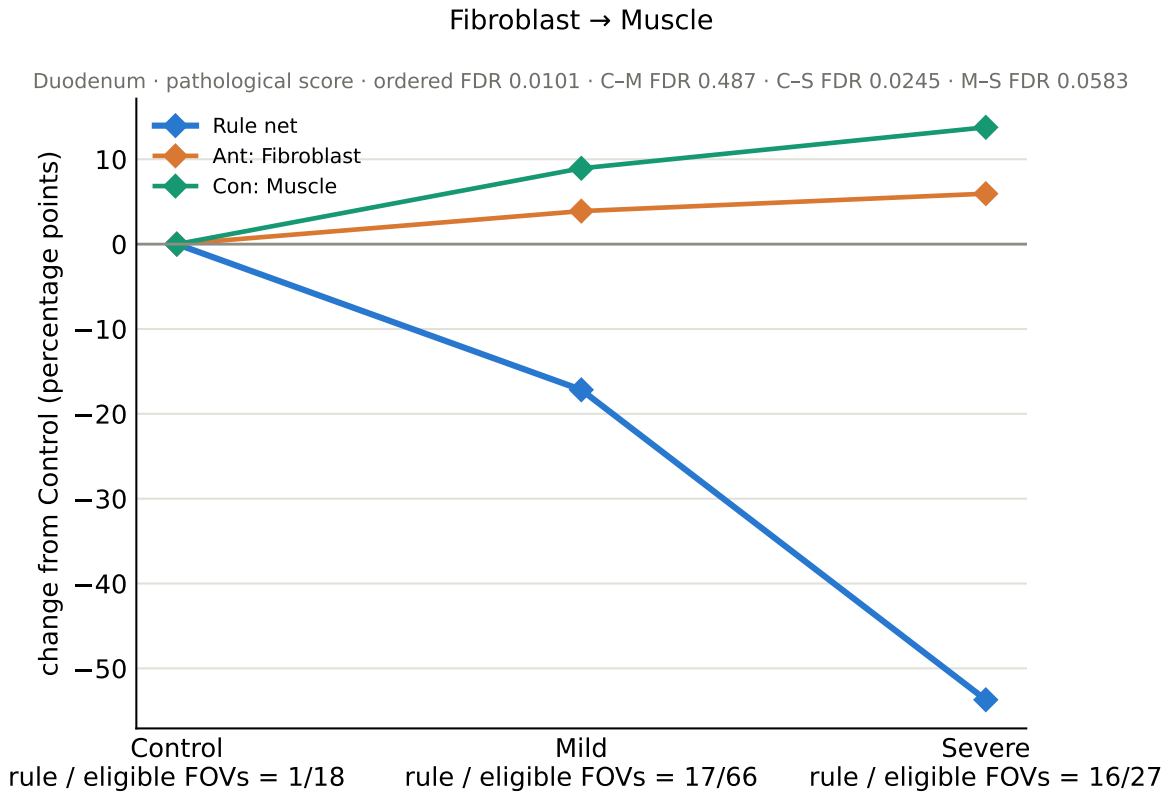


Figure 1: Blue is the change in Fibroblast → Muscle net score (attraction percentage minus avoidance percentage among eligible FOVs). Orange and green are changes in the mean within-FOV percentages of Fibroblast and Muscle cells, using those same FOVs. Every value is the stage mean minus the Control mean, in percentage points. The x-axis reports FOVs with a significant rule / eligible FOVs. Eligibility requires at least 20 cells of both types. See the [test guide above](#).

The stage-matched FOV examples make the same result spatially concrete: the upper row keeps the complete tissue context, while the lower row isolates Fibroblast and Muscle cells. These maps illustrate representative rule-bearing FOVs; the preceding aggregate plot is the evidence for the stage change.

Duodenum Fibroblast → Muscle — representative FOVs

pathological score · median-strength rule-bearing FOV in each stage · eligibility ≥ 20 cells/type

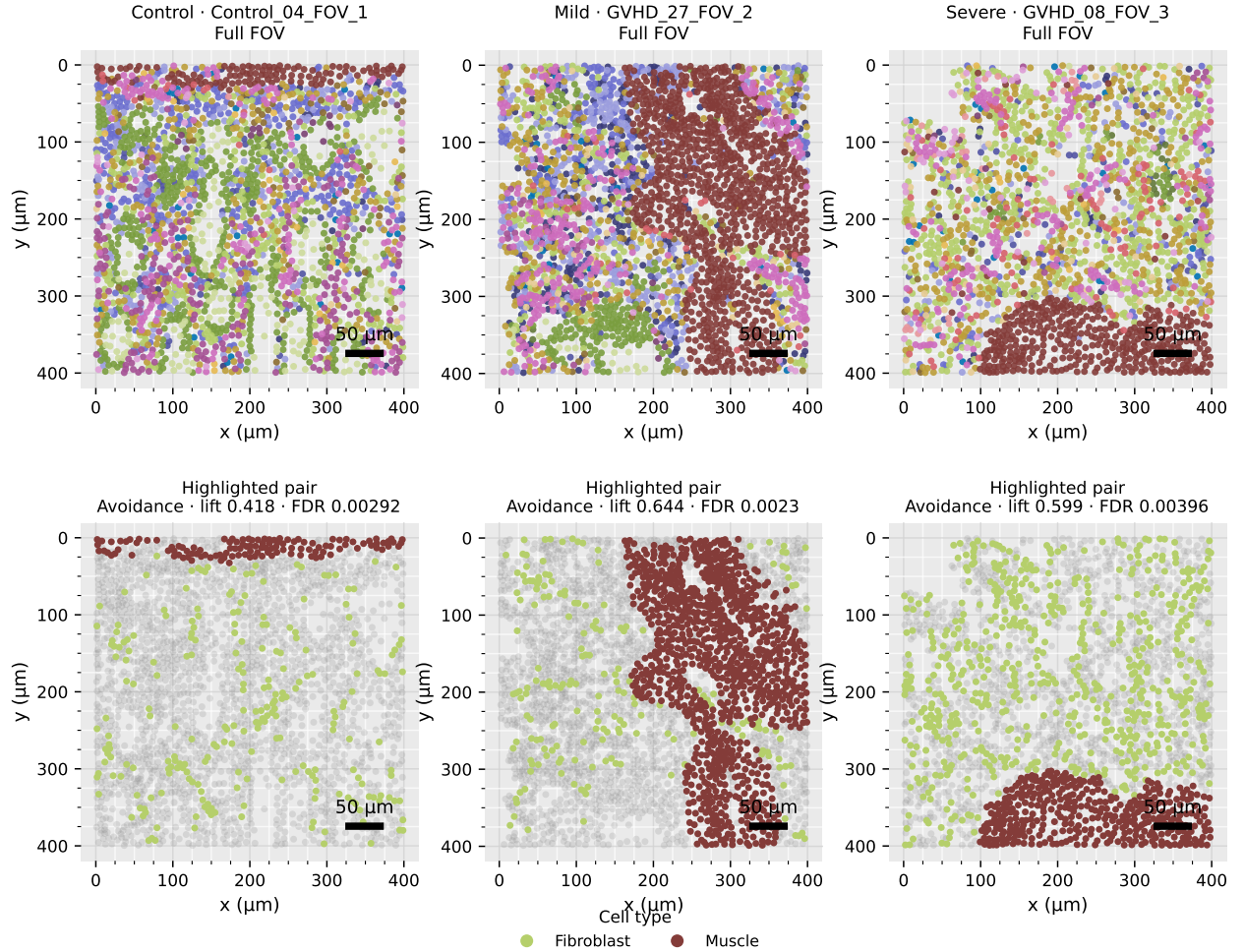


Figure 2: Within each pathological stage, the displayed FOV is selected automatically: first take the more frequent nonzero rule direction among eligible rule-bearing FOVs, then choose the occurrence whose Lift is closest to that stage's median. The upper row shows every cell; the lower row greys out cells outside the named pair. The state, Lift, and mining FDR printed above the lower map belong to that individual FOV, not to the between-stage test.

The same Fibroblast → Muscle separation appears when the biopsies are regrouped by clinical score, which makes it unlikely that the result is created by one severity definition.

Evidence strength: Strong.

Duodenum Fibroblast → Muscle beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

Fibroblast → Muscle

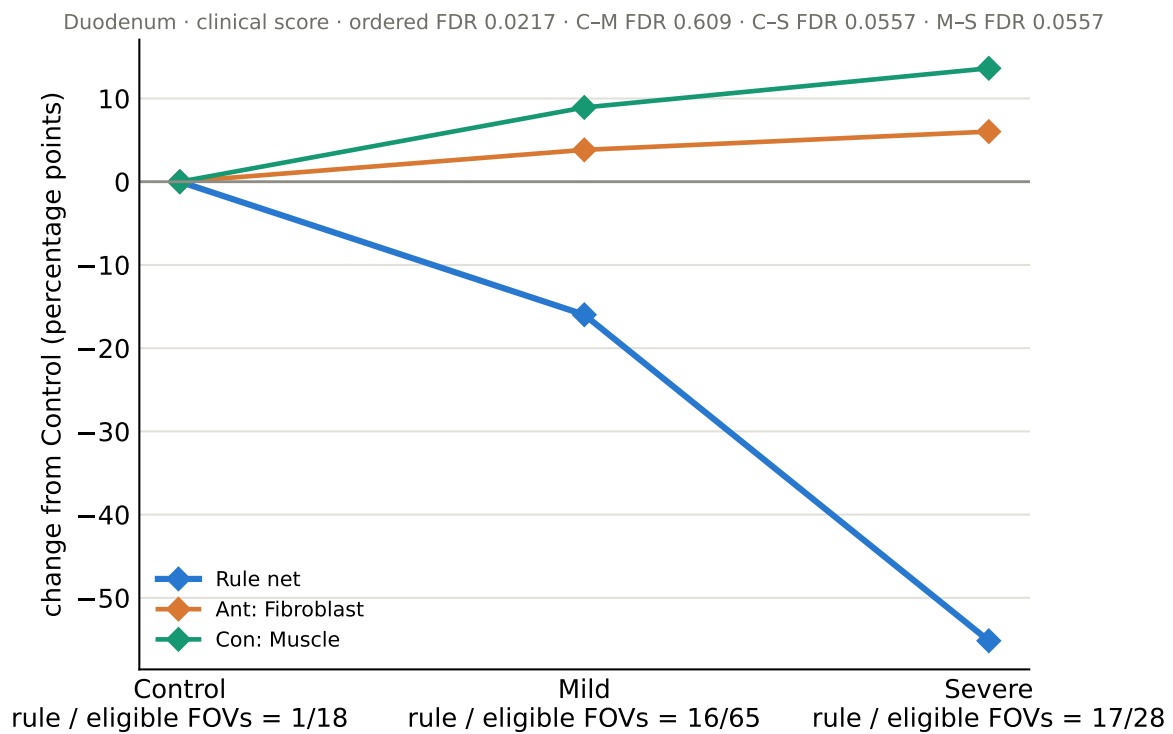


Figure 3: Values follow Figure 1; stages are defined by clinical score.

Direction is also retained rather than collapsing a cell pair into undirected co-occurrence: Colon CD4T → Goblet relaxes strongly with pathological severity, while the reverse rule starts more avoidant and changes by a different amount.

Evidence strength: Moderate.

Colon CD4T → Goblet beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

CD4T → Goblet

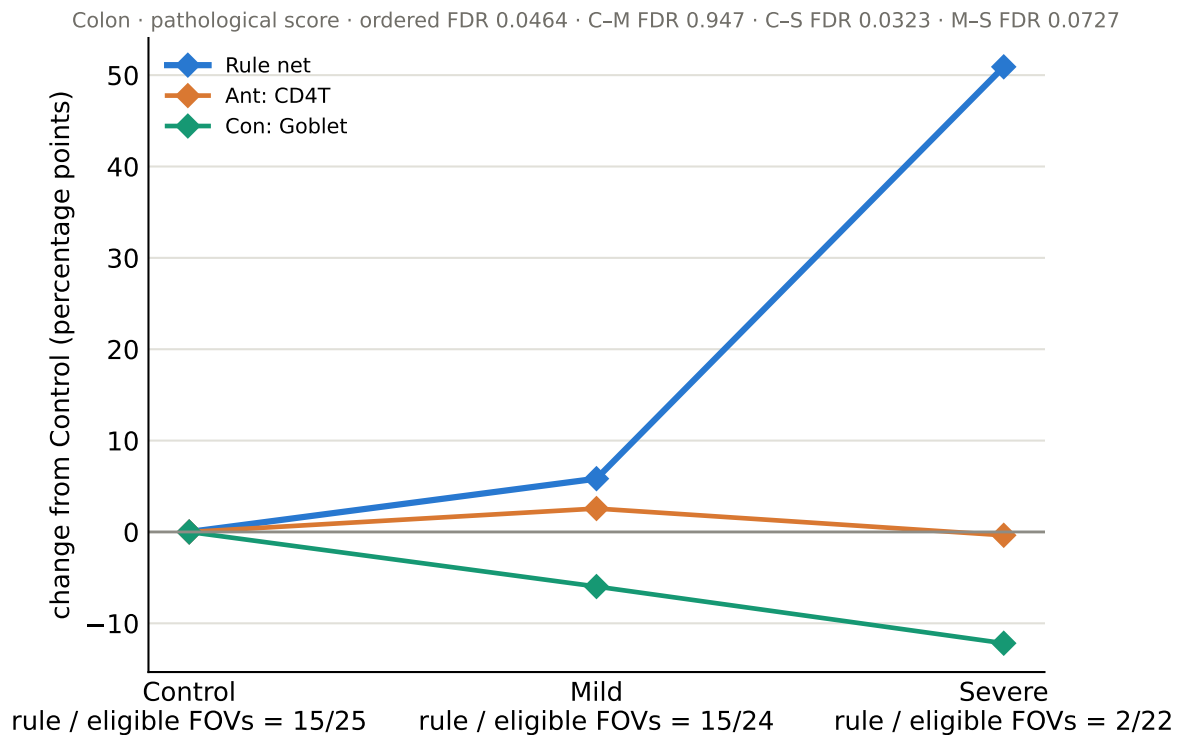


Figure 4: Values follow Figure 1; stages use pathological score in Colon.

Reading the reverse direction separately shows why it must remain a separate rule: Goblet → CD4T has its own baseline, trend, corrected tests, and abundance context.

Evidence strength: Moderate.

Colon Goblet → CD4T beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

Goblet → CD4T

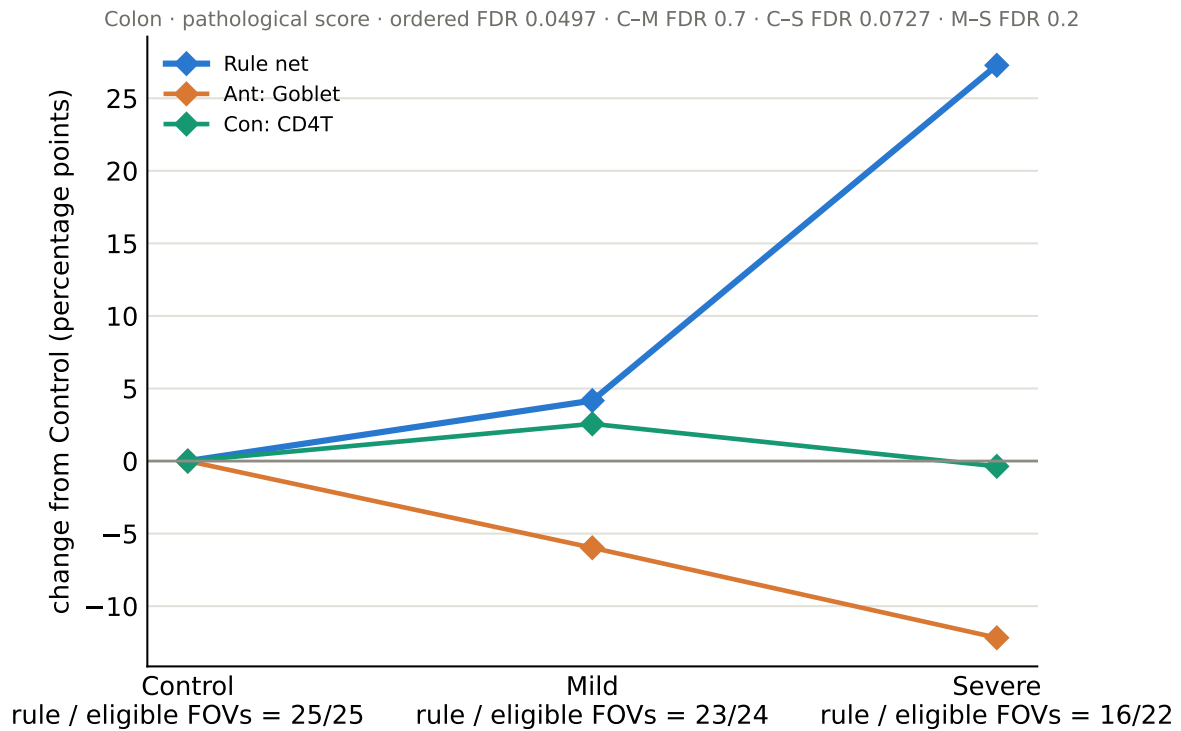


Figure 5: Values follow Figure 1; this reverse rule is calculated independently from CD4T → Goblet.

The method also preserves organ context. Goblet → Endothelial remains strongly avoidant in Colon but is much nearer neutral in Duodenum at matched clinical stages, and the lower panels show that this is not simply the same difference in cell abundance.

Evidence strength: Strong.

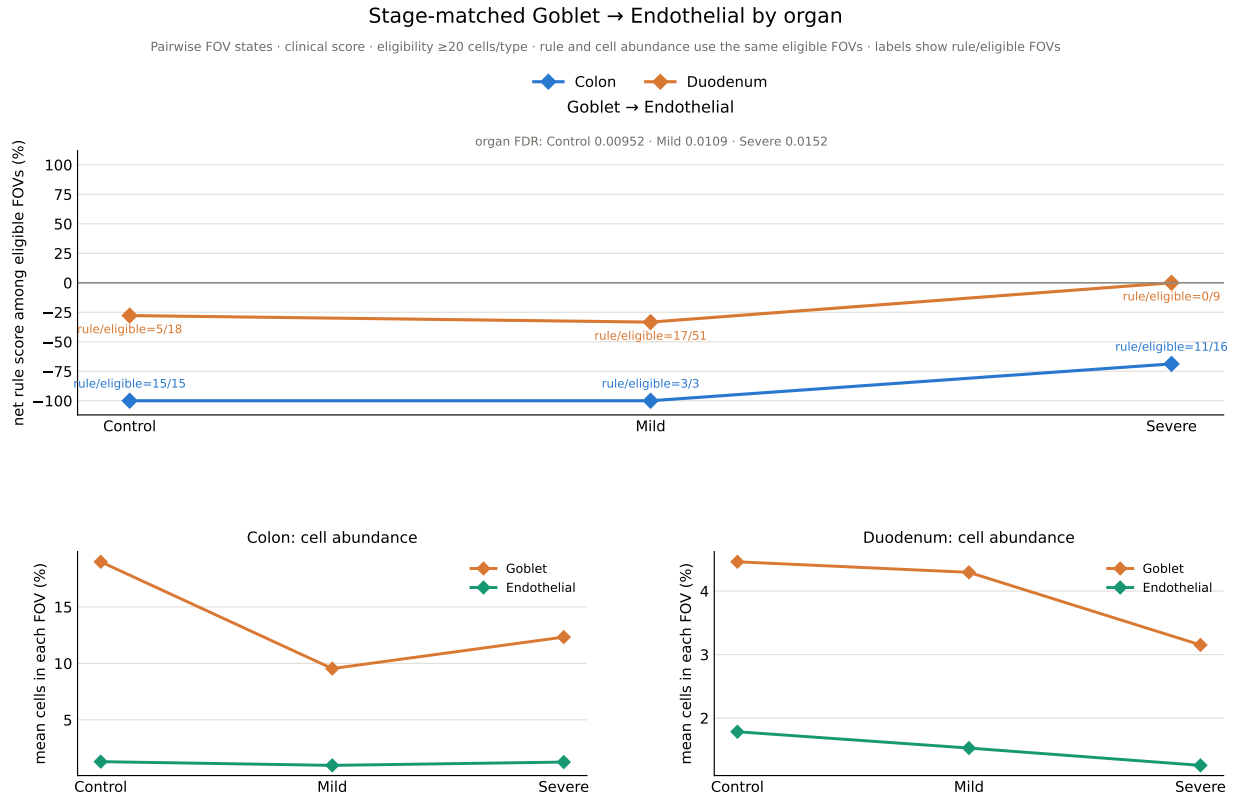


Figure 6: The upper panel gives the absolute Goblet → Endothelial net score (attraction percentage minus avoidance percentage among eligible FOVs) for each organ and stage. The lower panels give the absolute mean within-FOV percentages of Goblet and Endothelial cells, using the same eligible FOVs. Colon and Duodenum are compared within each clinical stage; organ FDR is corrected jointly across all tested rules and stages. Point labels report FOVs with a significant rule / eligible FOVs. See the [test guide above](#).

2 Rule-level alignment with the spatial-atlas paper

The strongest direct alignment is Paneth-cell loss. *The atlas reports “loss of Paneth cells.”* The count panel shows that Paneth eligibility collapses in Severe Duodenum, while the rule panel shows what remains measurable rather than confusing ineligibility with “no rule.”

Evidence strength: Strong for Paneth loss; moderate for the spatial rule.

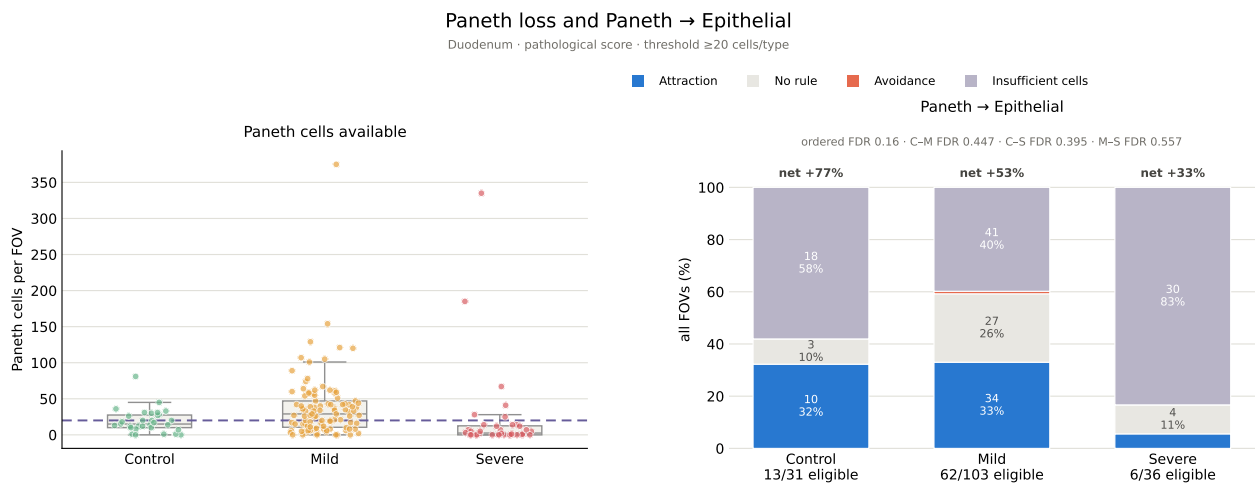


Figure 7: The left panel shows the raw number of Paneth cells in every FOV; the dashed line marks 20 cells. In the right stacked bars, each segment is the percentage of all stage FOVs classified as attraction, no rule, avoidance, or insufficient cells for Paneth → Epithelial. The printed net is attraction minus avoidance as a percentage of eligible FOVs. Both panels use pathological stages in Duodenum. See the [test guide above](#).

The representative fields put the Paneth result back into tissue context: they show typical rule-bearing FOVs where Paneth cells remain measurable, rather than selecting the most extreme examples.

Duodenum Paneth → Epithelial — representative FOVs

pathological score · median-strength rule-bearing FOV in each stage · eligibility ≥ 20 cells/type

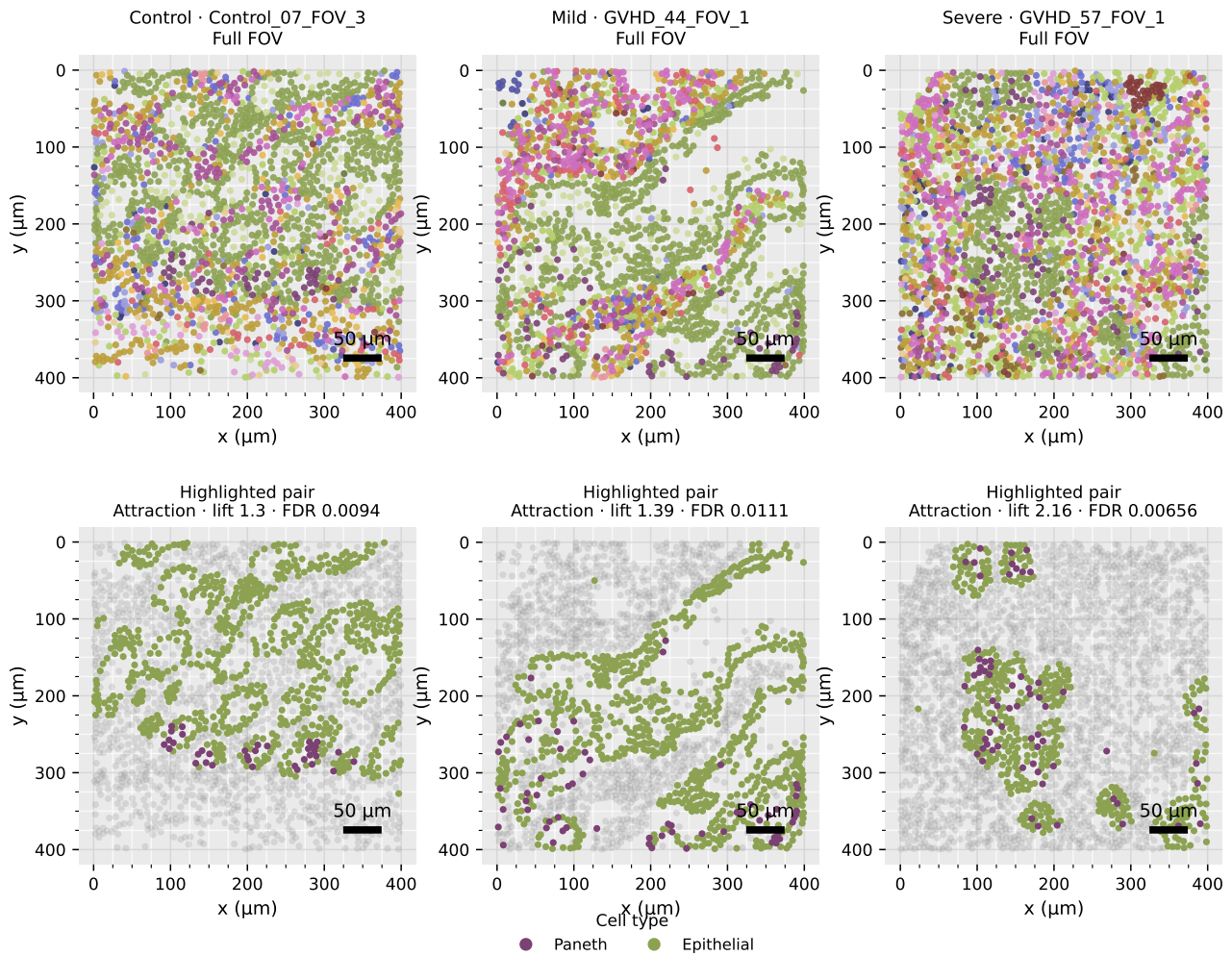


Figure 8: FOV selection and row meanings follow Figure 2; stages without a rule-bearing eligible FOV are stated explicitly.

The reverse direction adds a spatial hypothesis within the small Paneth-rich remainder: Epithelial → Paneth rises with severity, but only 6 Severe FOVs are eligible, so the figure supports a selected surviving niche rather than the whole Severe group.

Evidence strength: Moderate.

Paneth loss and Epithelial → Paneth

Duodenum · pathological score · threshold ≥ 20 cells/type

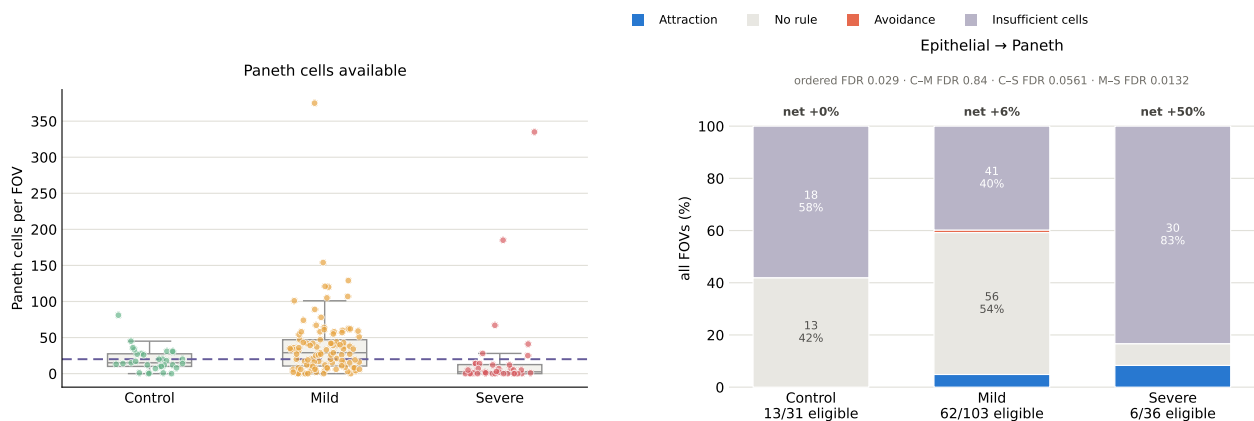


Figure 9: Values follow Figure 7; the reverse rule is calculated independently and “eligible” gives the FOV count passing the 20-cell threshold.

Endocrine accumulation is recovered just as directly. *The atlas reports “accumulation of endocrine cells in the crypts.”* No Control FOV reaches the eligibility threshold, whereas disease creates eligible FOVs with a predominantly attractive Endocrine → Epithelial rule; this is a clear qualitative translation, not a Control-to-disease significance test.

Evidence strength: Moderate.

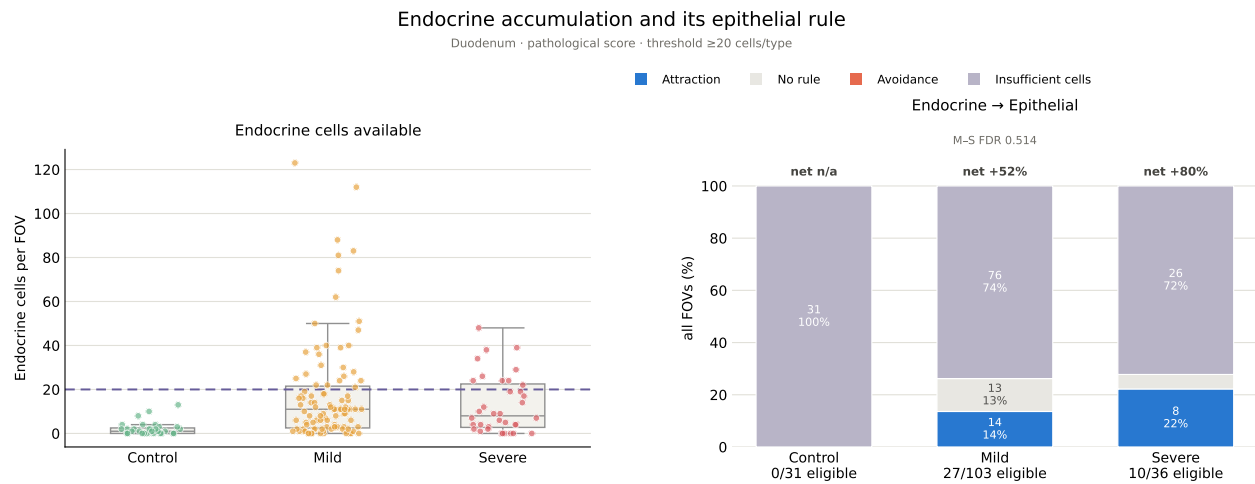


Figure 10: Values follow Figure 7; no Control FOV is eligible for Endocrine → Epithelial.

The tissue maps make the limitation visible as well as the signal: Control has no eligible Endocrine field, while the displayed disease examples are typical among the measurable rule-bearing FOVs.

Duodenum Endocrine → Epithelial — representative FOVs

pathological score · median-strength rule-bearing FOV in each stage · eligibility ≥ 20 cells/type

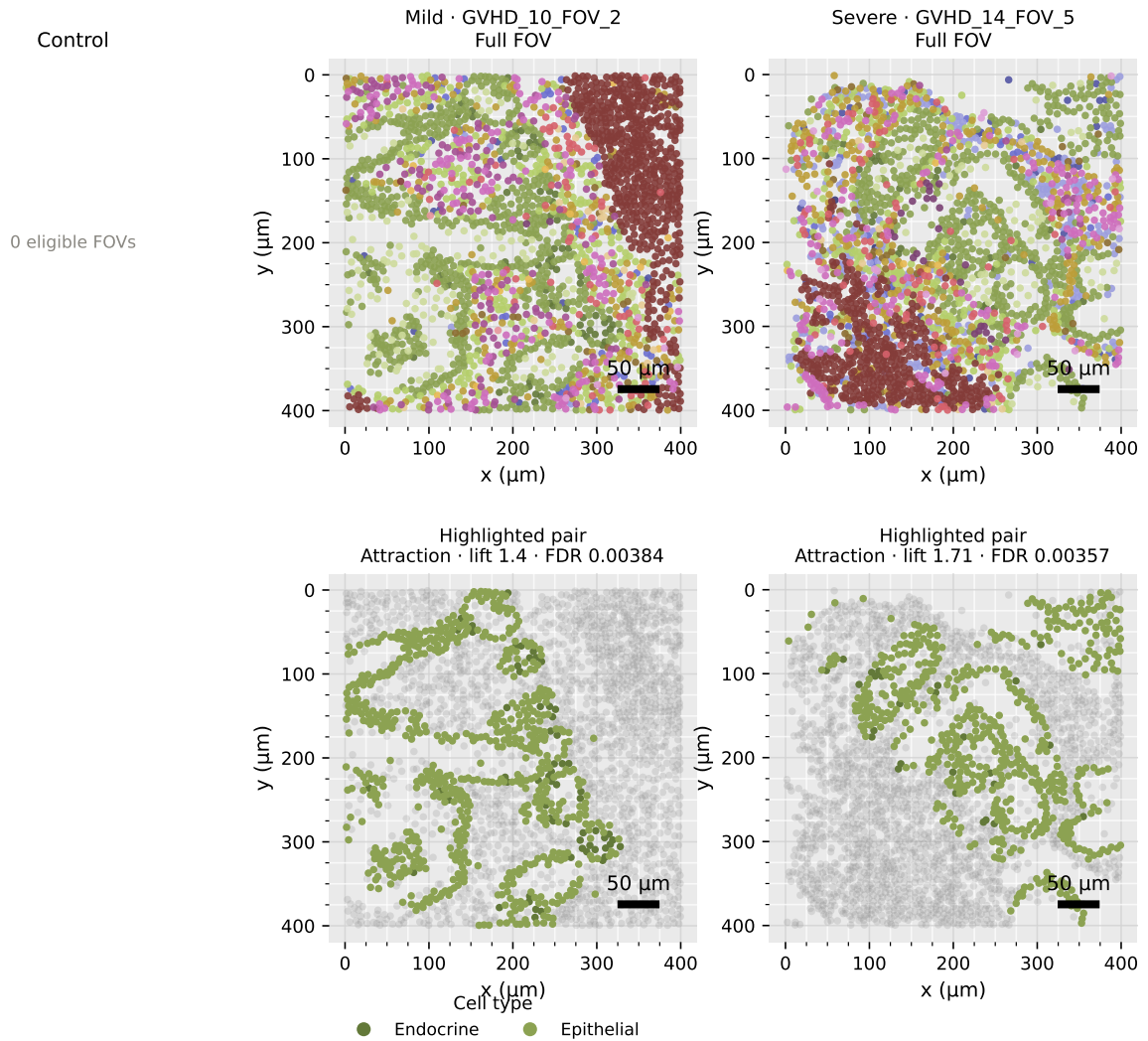


Figure 11: Selection follows Figure 2; “0 eligible FOVs” means that fewer than 20 cells of at least one required type were available in every FOV of that stage.

The atlas reports that “CD4 T cells, plasma cells, and macrophages were spatially organized from the crypts to the tip of the villus.” We therefore screened all six directions among these three cell types in both organs and both severity systems. Colon CD4T → Plasma is the strongest translation: its net score falls from +56% in Control to 0% in clinically Severe disease (C–S FDR 0.029), with the same loss under pathological staging (best FDR 0.016).

Evidence strength: Strong data signal; indirect alignment with the paper’s zonation result.

Colon CD4T → Plasma beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

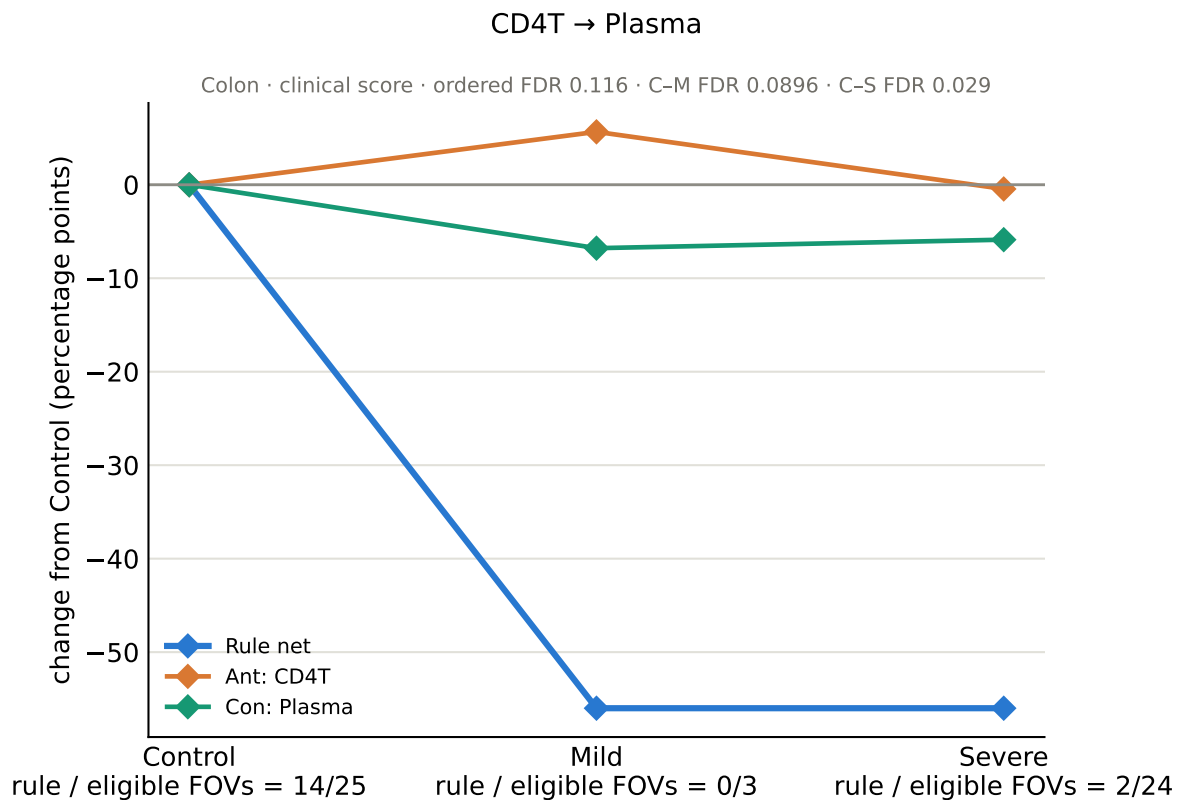


Figure 12: Values follow Figure 1; stages use clinical score in Colon.

Representative fields show the CD4T-Plasma context behind that aggregate change. They are chosen automatically near the median Lift of the dominant rule state and are illustrations, not additional statistical tests.

Colon CD4T → Plasma — representative FOVs

clinical score · median-strength rule-bearing FOV in each stage · eligibility ≥ 20 cells/type

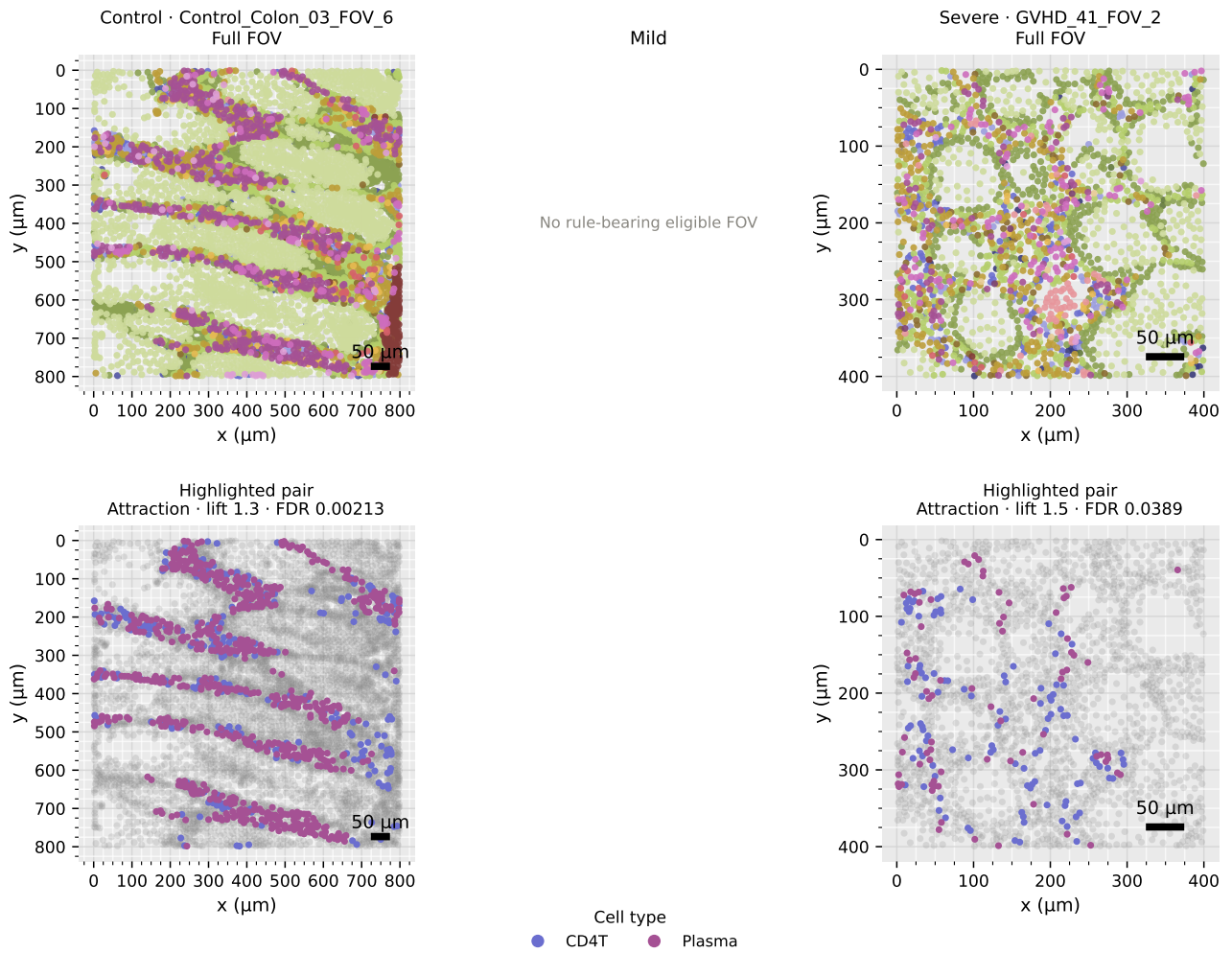


Figure 13: Selection and row meanings follow Figure 2; stages use clinical score in Colon.

The reverse Plasma → CD4T direction also weakens, from +48% in Control to +17% in clinically Severe disease. Its best FDR is 0.174, so it supports bidirectional loss of the Control organization visually but is not independently significant.

Evidence strength: Moderate-to-weak.

Colon Plasma → CD4T beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

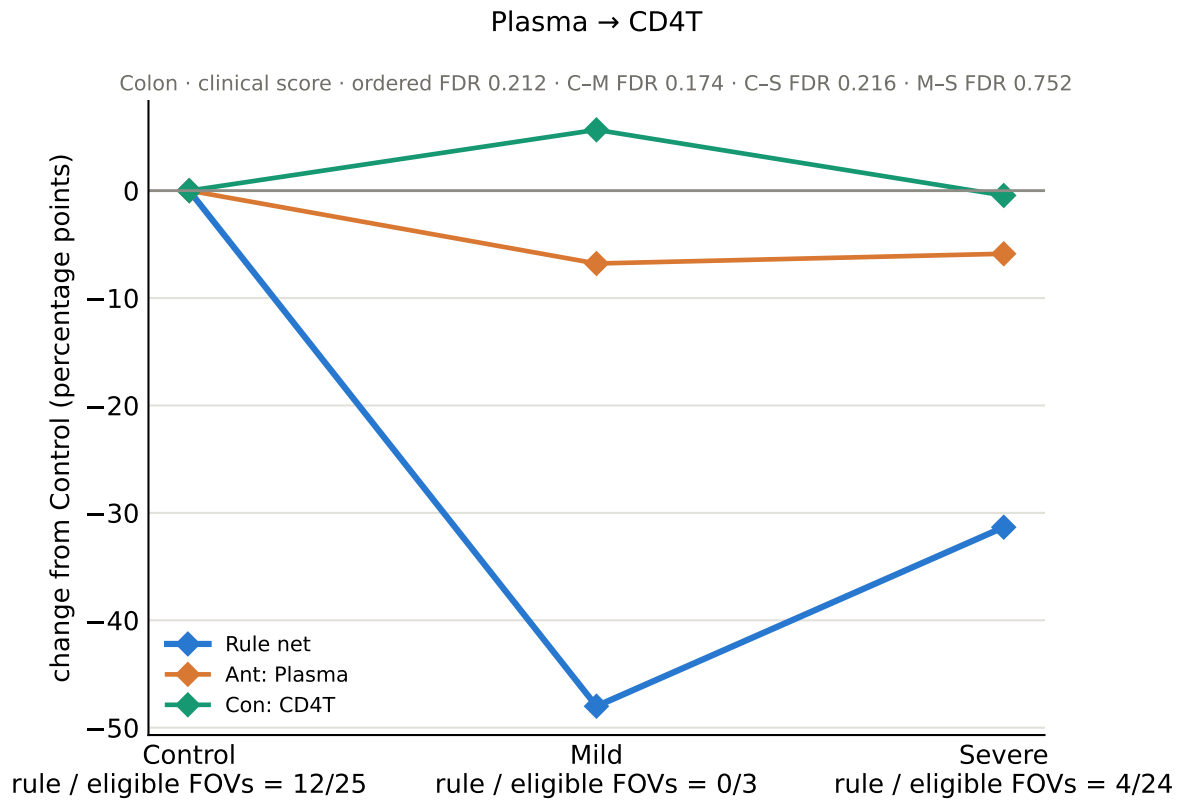


Figure 14: Values follow Figure 1; this reverse direction is tested separately using clinical stages in Colon.

Macrophage → CD4T provides a second component of the same Colon pattern: the pathological net score falls from +48% to +12% (best FDR 0.073), with a compatible but less regular clinical result. This is suggestive rather than conventionally significant.

Evidence strength: Moderate.

Colon Macrophage → CD4T beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

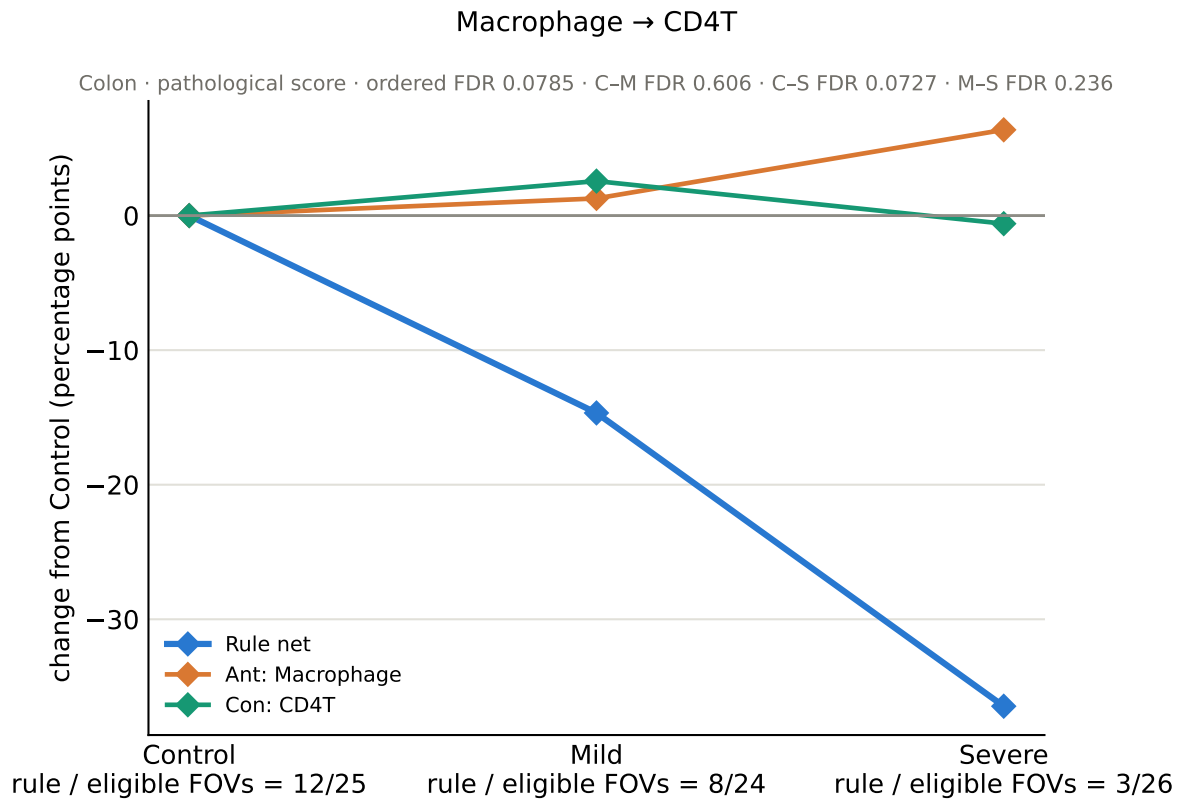


Figure 15: Values follow Figure 1; stages use pathological score in Colon.

The representative fields show the macrophage-CD4T pair within complete Colon tissue and then in isolation, making the aggregate loss of attraction spatially inspectable.

Colon Macrophage → CD4T — representative FOVs

pathological score · median-strength rule-bearing FOV in each stage · eligibility ≥ 20 cells/type

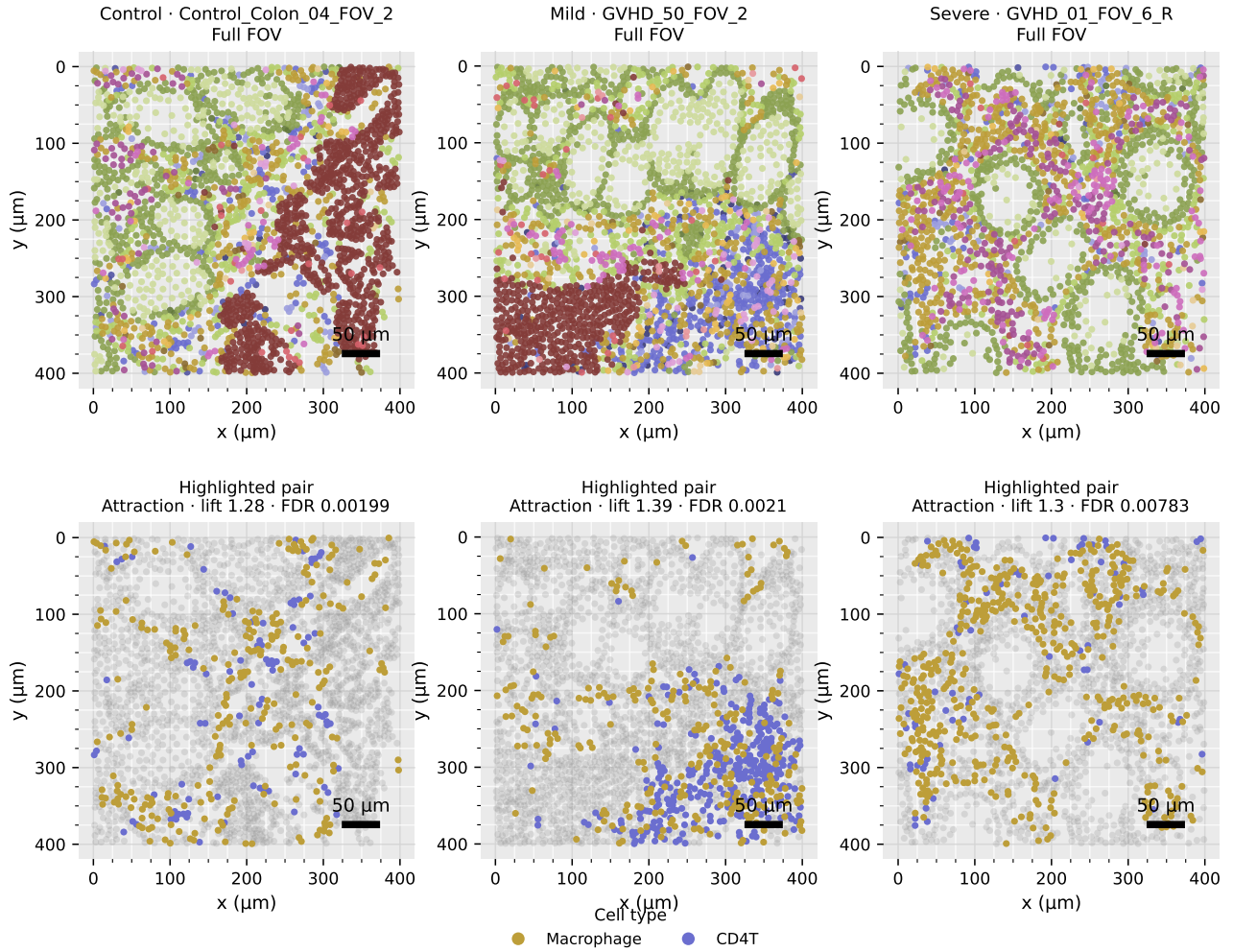


Figure 16: Selection and row meanings follow Figure 2; these maps do not establish the paper’s patient-level macrophage subgroup.

No Duodenal direction in the CD4T–Plasma–Macrophage family reached FDR 0.10 or a 30-percentage-point Control–Severe change. This does not reproduce the paper’s exact crypt–villus zonation, but it prevents a weak Duodenal substitute from being presented as alignment.

The atlas also reports “marked inpatient heterogeneity in the prevalence of CD8 T cells and neutrophils in the vicinity of individual crypts and villi.” In Duodenum, CD8T → Goblet becomes much more avoidant, from -21% in Control to -75% in pathologically Severe disease (C–S FDR 0.013). This is a strong rule change, although Goblet is only a proxy for the paper’s crypt-level epithelial context.

Evidence strength: Strong data signal; indirect paper alignment.

Duodenum CD8T → Goblet beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

CD8T → Goblet

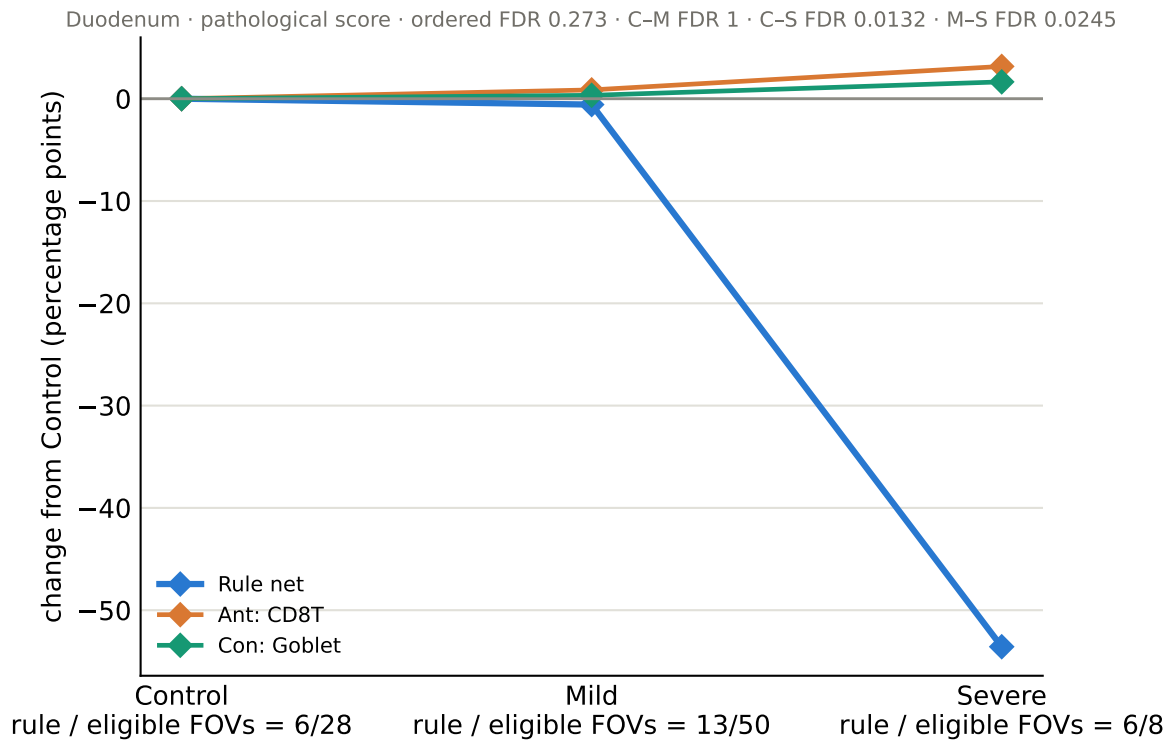


Figure 17: Values follow Figure 1; stages use pathological score in Duodenum, and these are FOV-level rather than crypt- or patient-level values.

The corresponding fields reveal what the summary represents spatially while preserving the surrounding tissue; they do not identify the paper's CD8-enriched patient subset.

Duodenum CD8T → Goblet — representative FOVs

pathological score · median-strength rule-bearing FOV in each stage · eligibility ≥ 20 cells/type

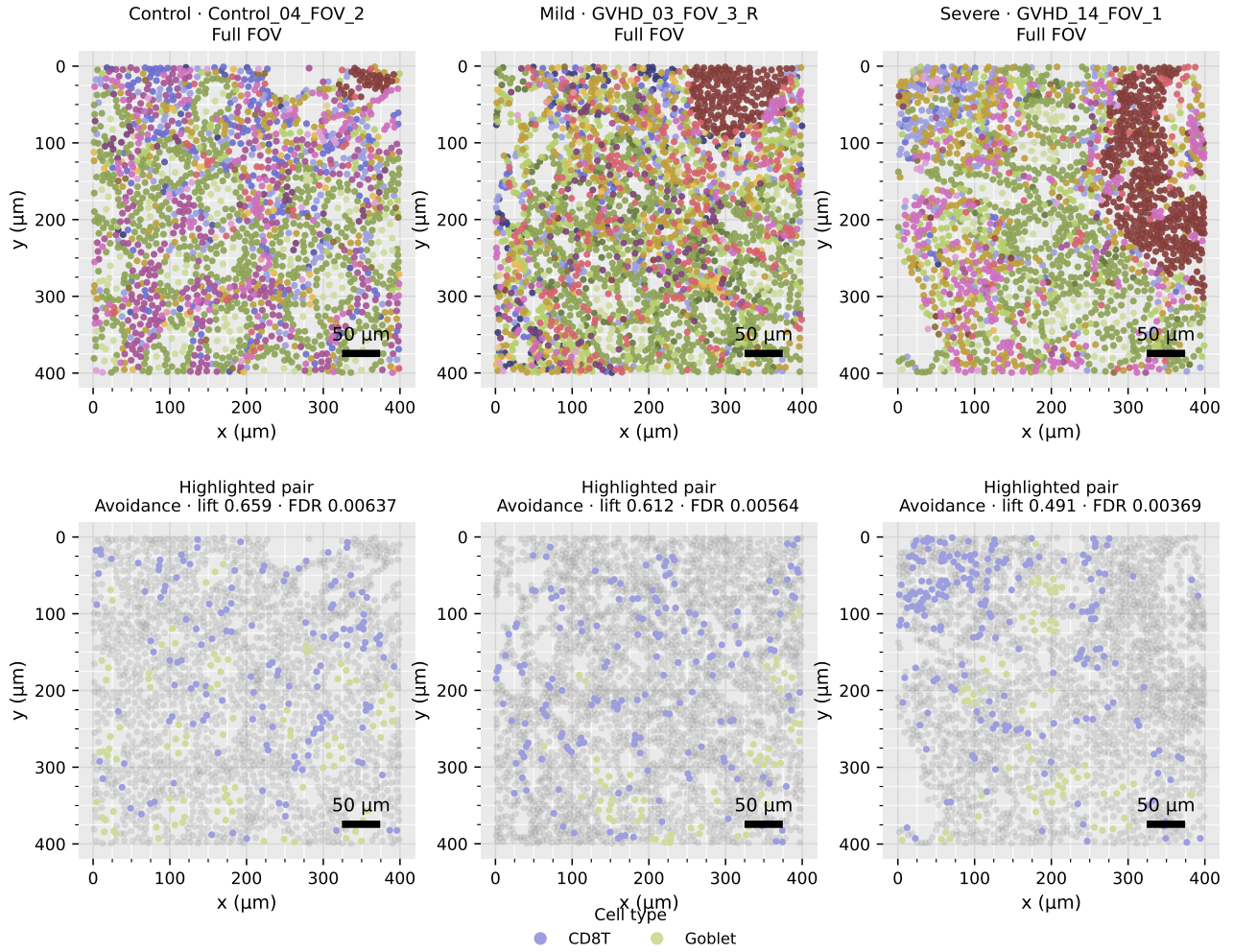


Figure 18: Selection and row meanings follow Figure 2; stages use pathological score in Duodenum.

The reverse Goblet → CD8T rule changes independently from -46% to -100% ; its best corrected FDR is 0.056. The similar direction under clinical staging supports the pattern, but the corrected evidence remains borderline.

Evidence strength: Moderate; borderline corrected evidence.

Duodenum Goblet → CD8T beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

Goblet → CD8T

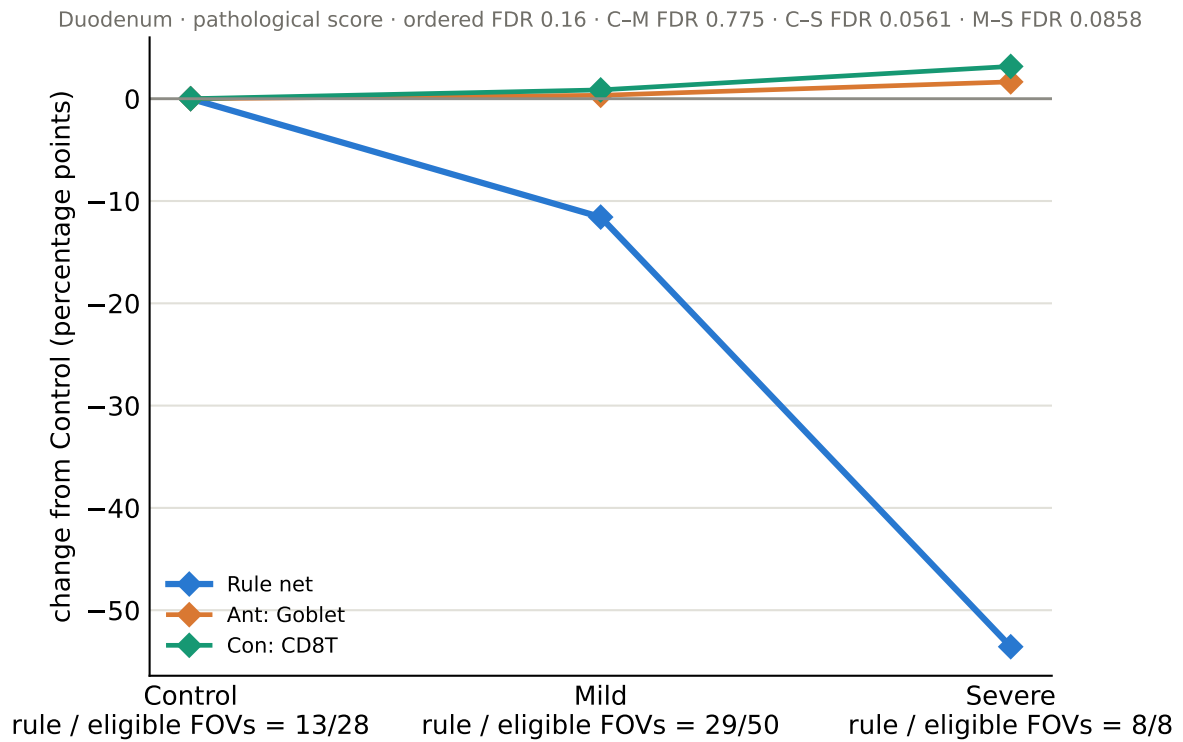


Figure 19: Values follow Figure 1; the reverse direction is calculated independently using pathological stages in Duodenum.

Representative reverse-rule fields make that directional calculation inspectable rather than implying that the two arrows are interchangeable.

Duodenum Goblet → CD8T — representative FOVs

pathological score · median-strength rule-bearing FOV in each stage · eligibility ≥ 20 cells/type

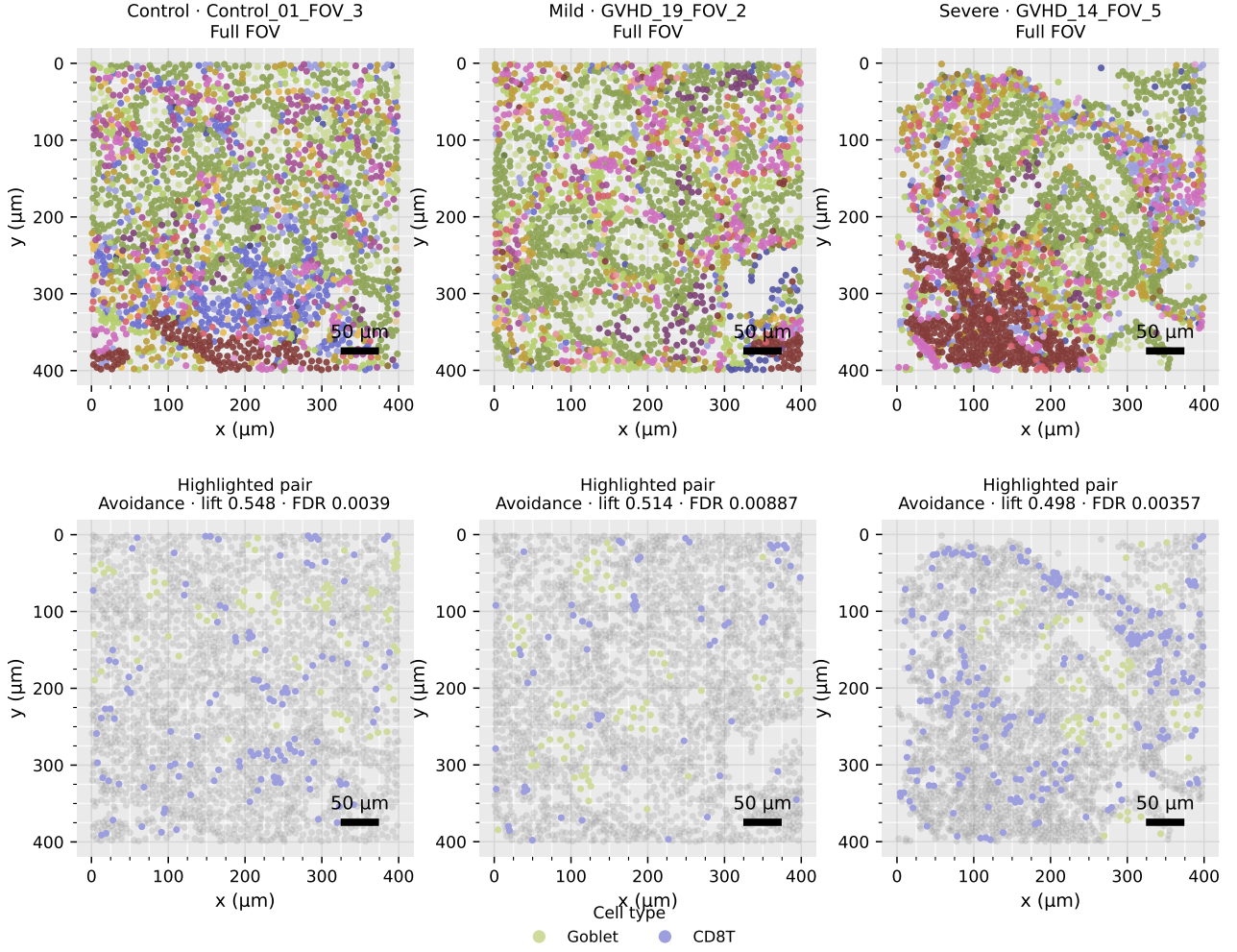


Figure 20: Selection and row meanings follow Figure 2.

The systematic screen found no meaningful severity change for CD8T $\leftrightarrow$ Epithelial in either organ, and Colon CD8T $\leftrightarrow$ Goblet was nearly stable from Control to Severe. Those negative results remain in the notebook audit table but are not shown here as uninformative figures. Pair rules also do not recover the paper's patient-level macrophage/neutrophil clusters; that requires a separate patient-state analysis.

3 New biologically interpretable findings

Duodenal Macrophage $\rightarrow$ Muscle becomes markedly more avoidant with pathological severity even though neither cell population disappears. This is biologically interesting because muscularis macrophages are known to participate in gut neuromuscular homeostasis (Müller et al., 2014), but the present result does not identify that macrophage subtype.

Evidence strength: Strong data signal; biologically exploratory.

Duodenum Macrophage → Muscle beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

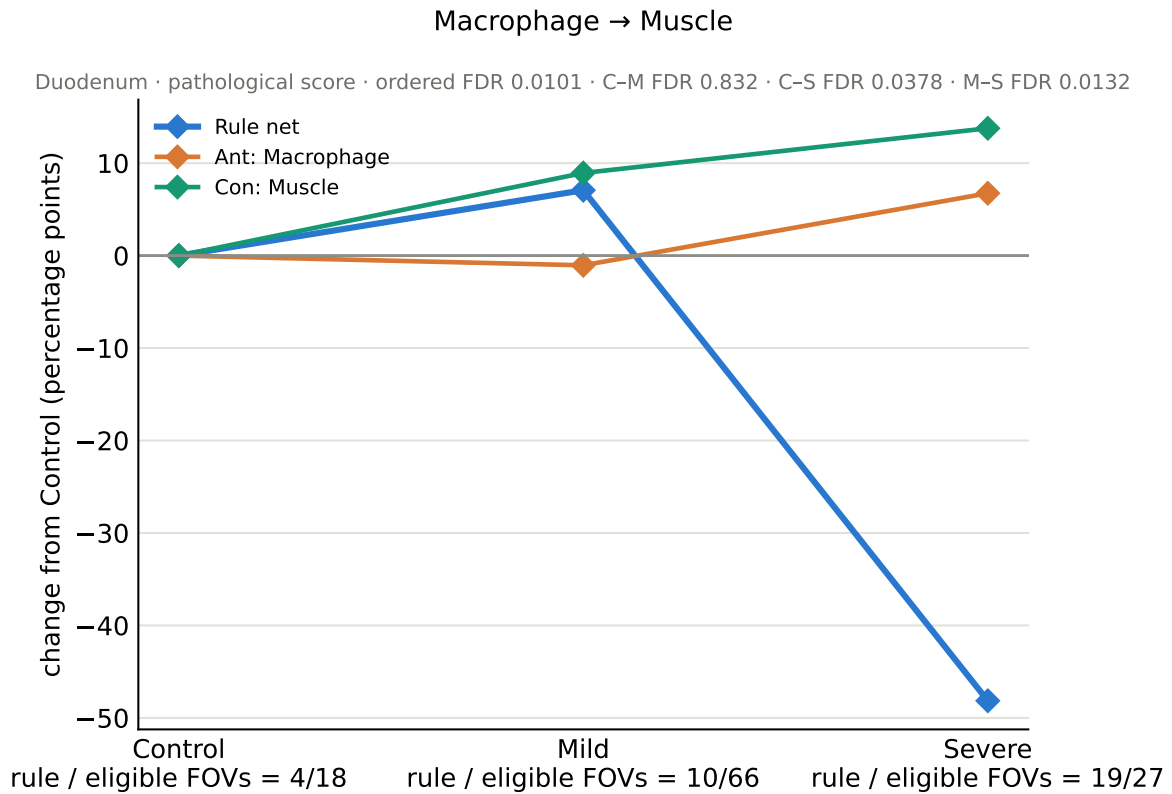


Figure 21: Values follow Figure 1; stages use pathological score in Duodenum.

Duodenal CD4T → Goblet becomes substantially less avoidant under pathological staging, while the cell-share trajectories are much smaller than the rule movement; this suggests altered immune access to the epithelial compartment rather than composition alone.

Evidence strength: Strong.

Duodenum CD4T → Goblet beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

CD4T → Goblet

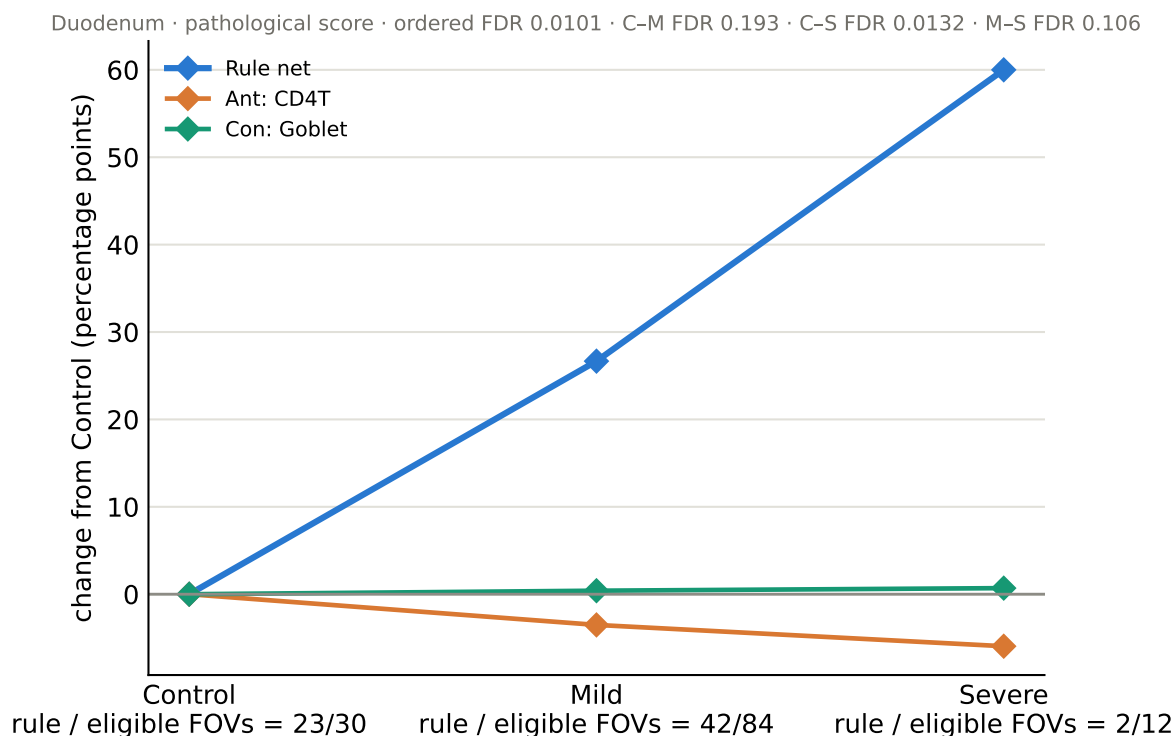


Figure 22: Values follow Figure 1; stages use pathological score in Duodenum.

Representative FOVs show the tissue context behind that CD4T → Goblet summary, with the full field directly above the highlighted cell pair in each pathological stage; they are illustrations of the automatically detected trend, not separately selected proof.

Duodenum CD4T → Goblet — representative FOVs

pathological score · median-strength rule-bearing FOV in each stage · eligibility ≥ 20 cells/type

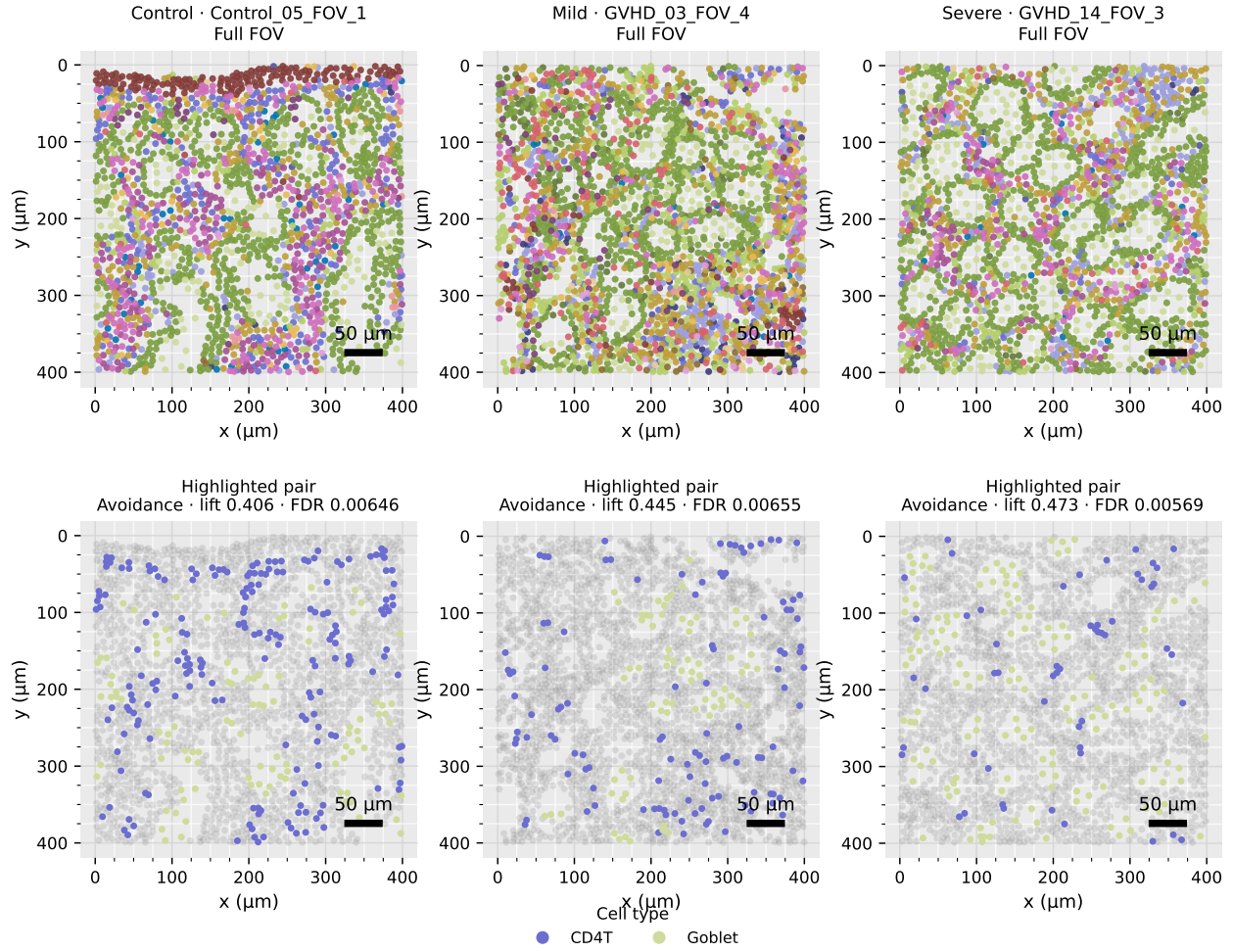


Figure 23: FOV selection and row meanings follow Figure 2; the state, Lift and rule-level FDR printed above each highlighted map belong to that individual FOV.

The clinical grouping independently preserves the same CD4T → Goblet direction, which strengthens the observation because the two severity definitions assign biopsies differently.

Evidence strength: Strong.

Duodenum CD4T → Goblet beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

CD4T → Goblet

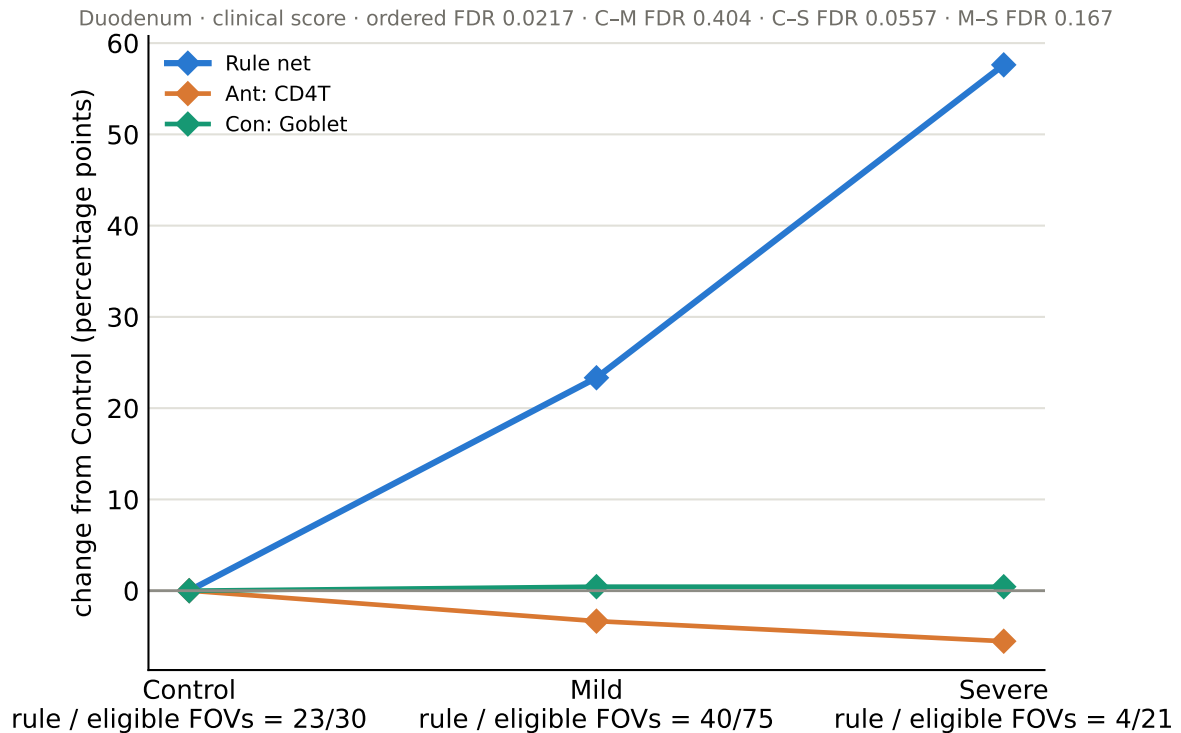


Figure 24: Values follow Figure 1; here biopsies are grouped by clinical rather than pathological score.

Colon Goblet → Muscle is strongly avoidant in Control and relaxes in disease; the matched cell shares do not reproduce the blue trajectory, making this a candidate change in epithelial-stromal organization.

Evidence strength: Strong.

Colon Goblet → Muscle beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

Goblet → Muscle

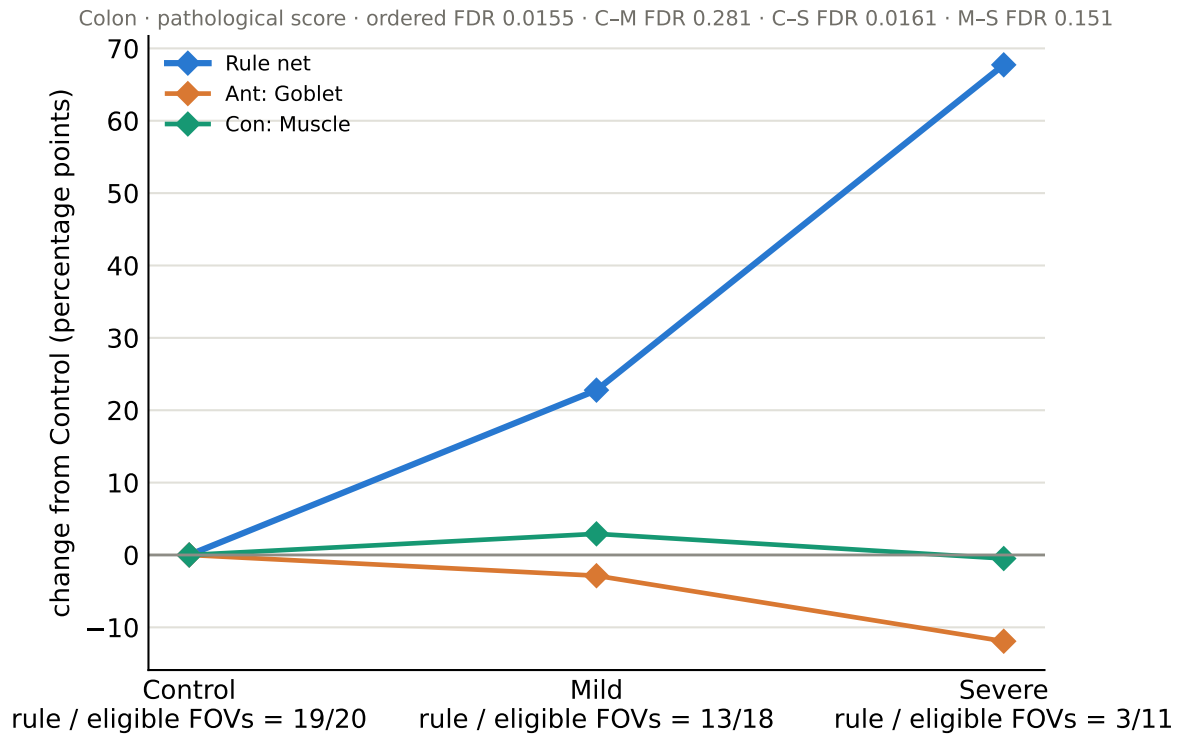


Figure 25: Values follow Figure 1; stages use pathological score in Colon.

Muscle → Goblet changes in the same broad direction but has independently estimated states and effect sizes, supporting a bidirectional reorganization rather than using the reverse arrow as a duplicate.

Evidence strength: Strong.

Colon Muscle → Goblet beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

Muscle → Goblet

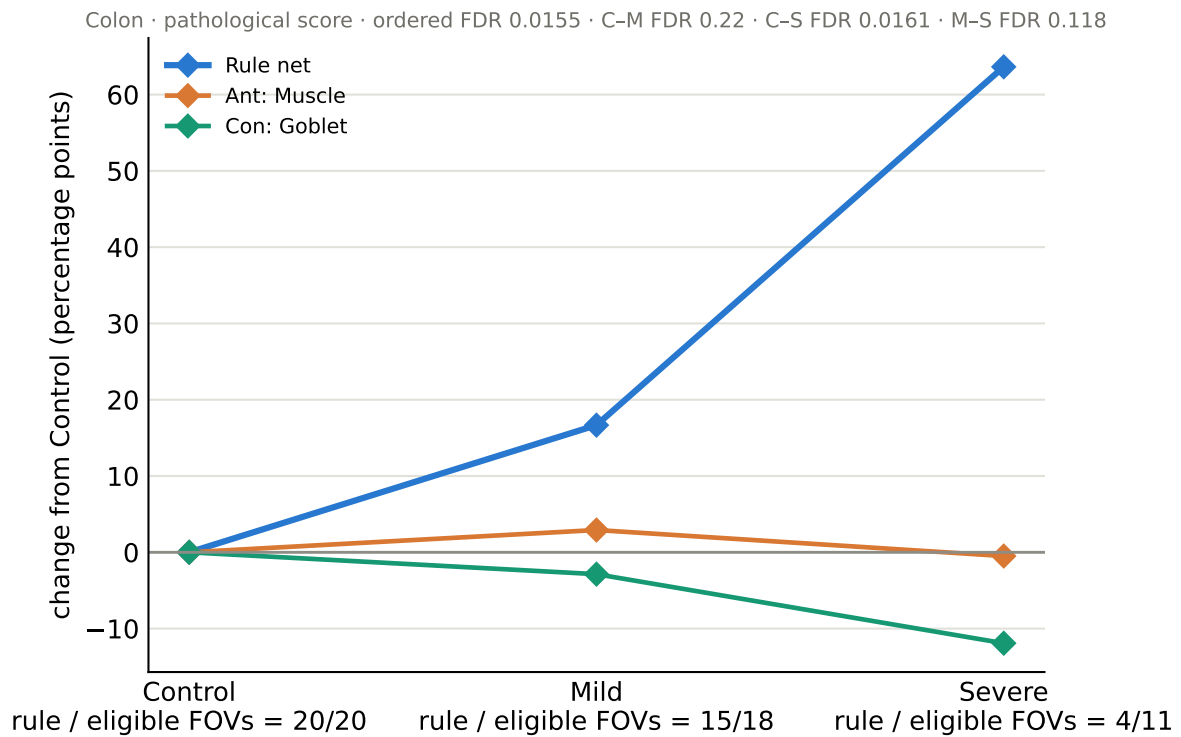


Figure 26: Values follow Figure 1; the reverse rule is tested independently under pathological staging in Colon.

The same Goblet → Muscle rule is also organ-specific at matched clinical stages: Colon is strongly avoidant while Duodenum stays near neutral, and both cell-abundance profiles are visible below the rule comparison.

Evidence strength: Strong.

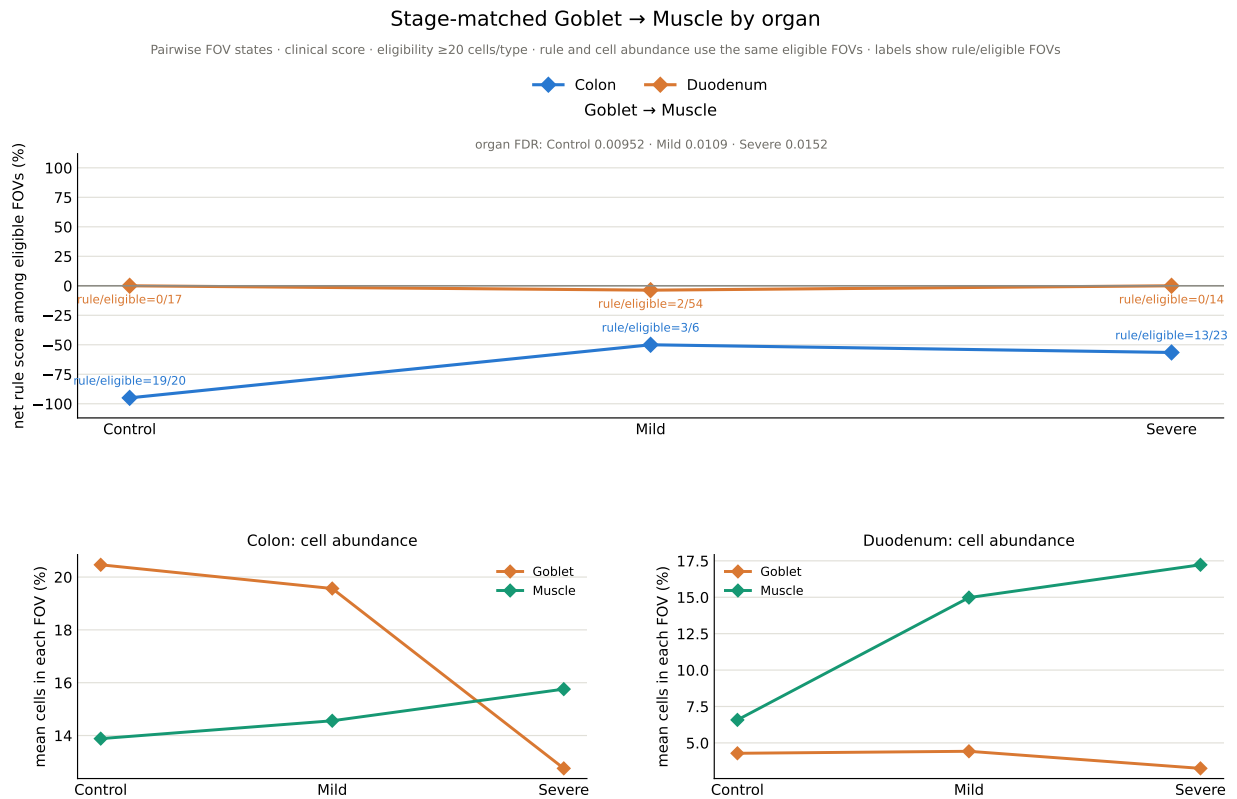


Figure 27: Values and “n” follow Figure 6; clinical stages are matched between organs, and the organ difference passes FDR 0.05 in at least two stages.

Fibroblast → Goblet shows the opposite organ ordering in disease, becoming more avoidant in Duodenum than Colon at Mild and Severe; the result is interesting because the organ gap persists after using rule-specific eligible FOVs for the abundance context.

Evidence strength: Moderate.

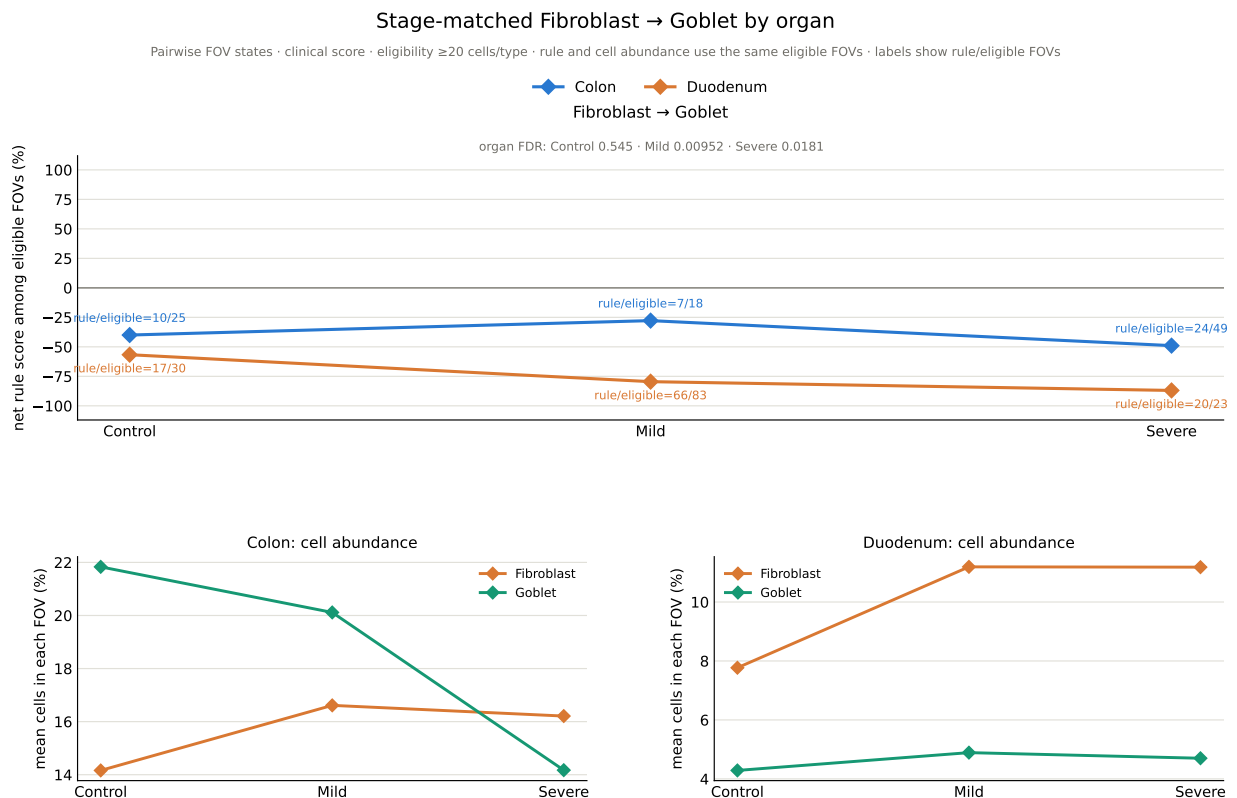


Figure 28: Values and “n” follow Figure 6; clinical stages are matched between organs.

Colon Goblet → SMV is nearly maximally avoidant in Control and Mild but relaxes in Severe while Goblet and SMV abundance move differently; this makes it a strong numerical lead for epithelial–vascular reorganization, although the atlas did not state this rule.

Evidence strength: Strong data signal; biologically exploratory.

Colon Goblet → SMV beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥20 cells/type

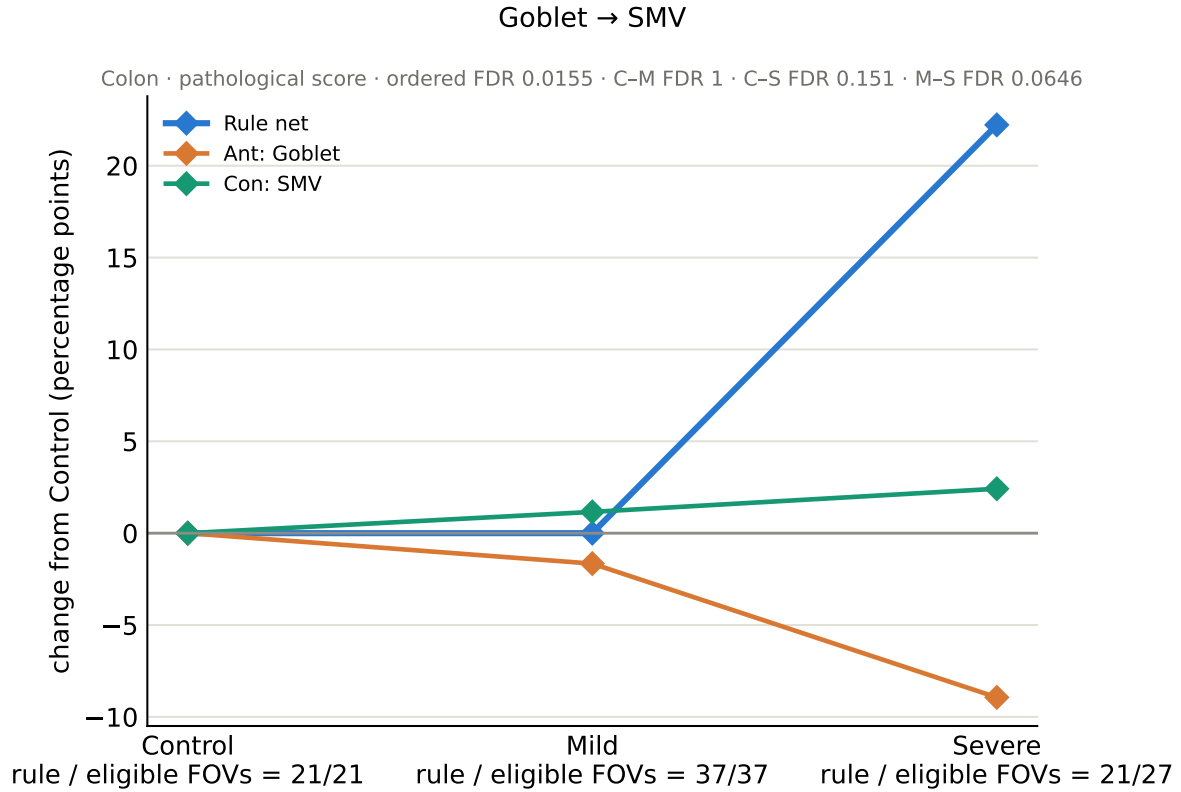


Figure 29: Values follow Figure 1; stages use pathological score in Colon.

Colon Fibroblast → Plasma moves from attraction toward neutrality, a plausible loss of an immune–stromal niche that complements the paper’s plasma-cell depletion; it is kept as a hypothesis because the ordered trend is borderline and the atlas gives no direct spatial counterpart.

Evidence strength: Moderate.

Colon Fibroblast → Plasma beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

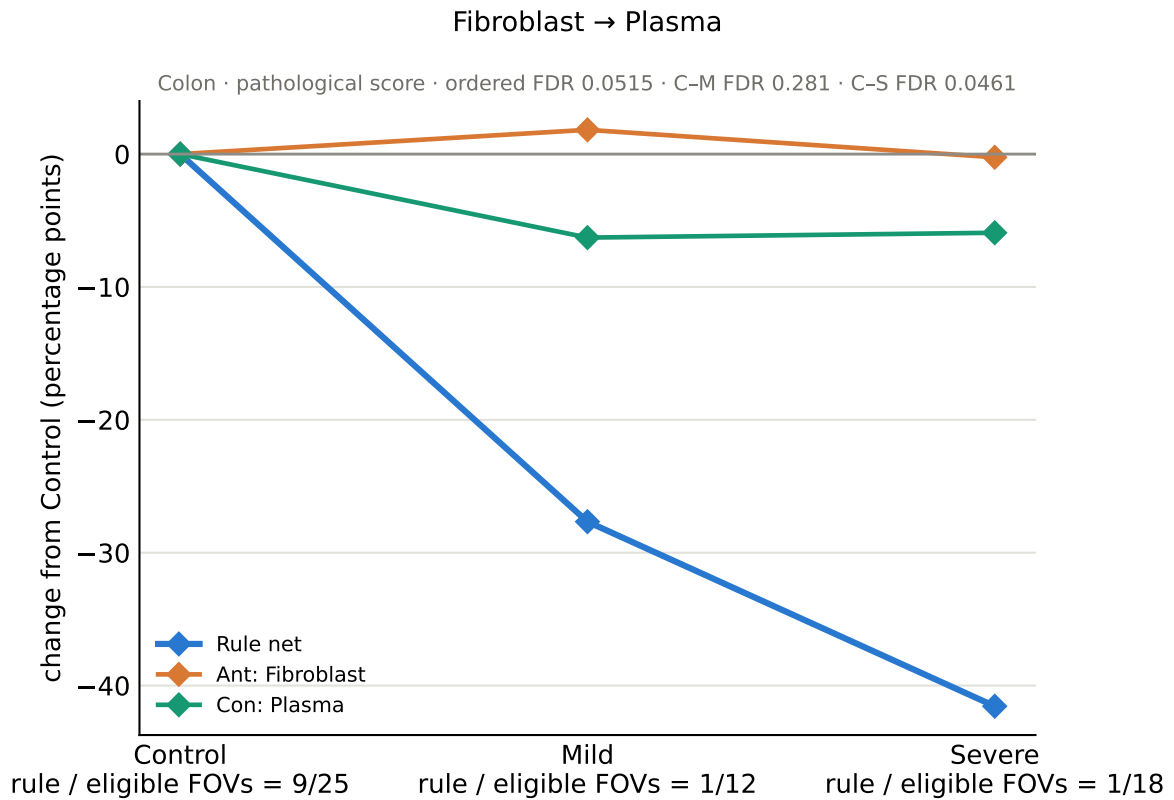


Figure 30: Values follow Figure 1; stages use pathological score in Colon. The Control–Severe contrast passes FDR 0.05, whereas ordered-trend FDR is approximately 0.052.

The remaining three results are scientifically interesting precisely because the plots make their limitation visible: each concerns a rare-cell niche and therefore has a very small eligible endpoint group. Paneth → Mast becomes strongly avoidant, but only three Severe FOVs support the estimate.

Evidence strength: Exploratory.

Duodenum Paneth → Mast beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥ 20 cells/type

Paneth → Mast

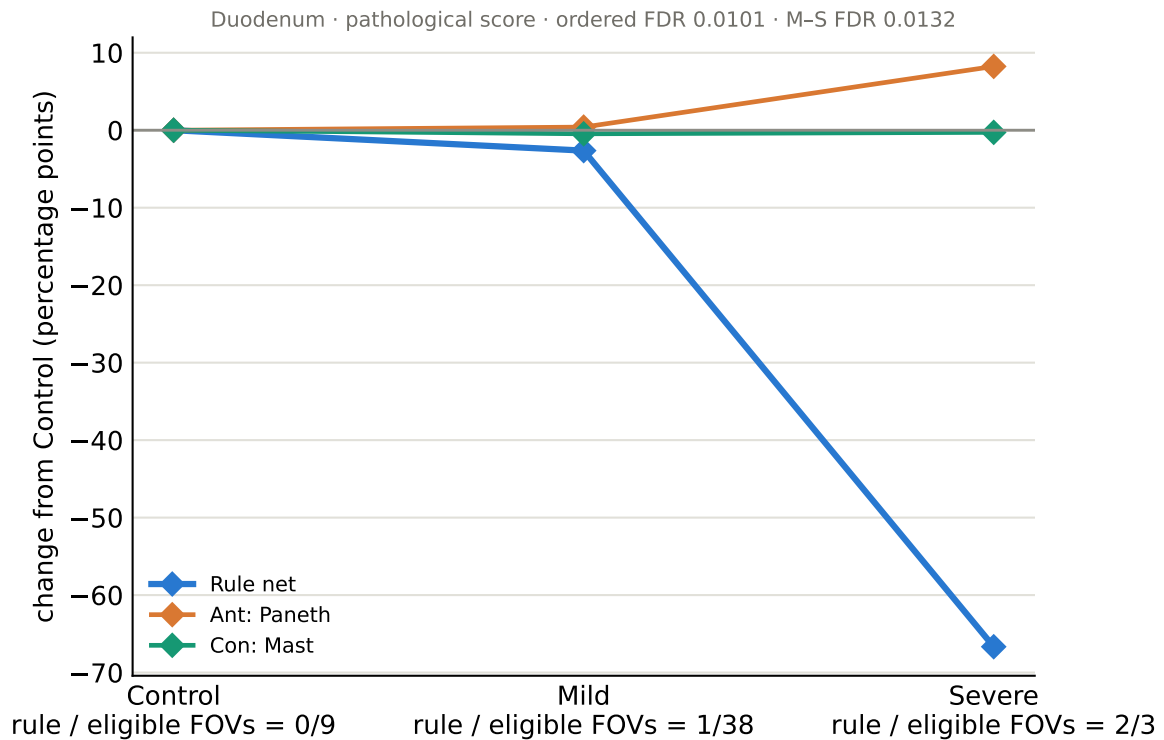


Figure 31: Values and “rule n ” follow Figure 1; stages use pathological score in Duodenum.

Paneth → Plasma_CD38 also becomes avoidant and passes corrected comparisons, which could mark reorganization of the surviving Paneth-rich niche, but only five Severe FOVs contain enough of both cell types.

Evidence strength: Exploratory.

Duodenum Paneth → Plasma_CD38 beside cell abundance

Each line starts at 0 in Control · rule and cell shares use the same eligible FOVs · eligibility ≥20 cells/type

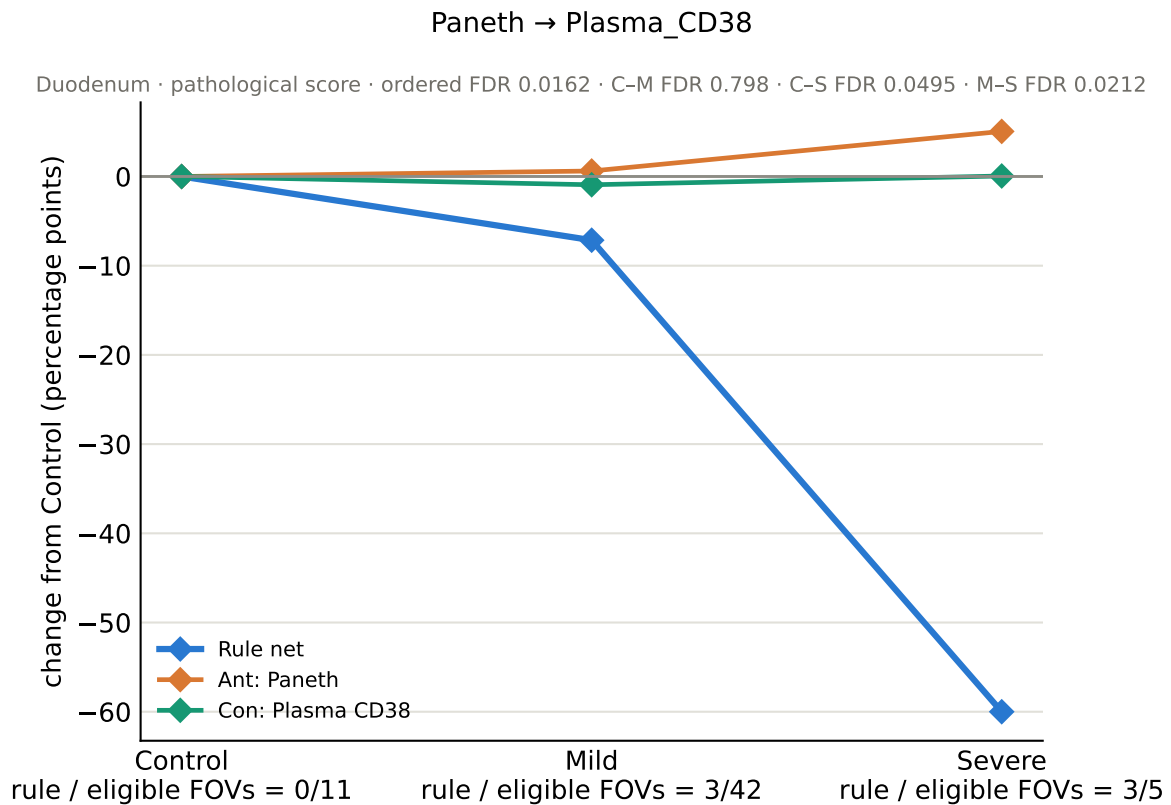


Figure 32: Values follow Figure 1; stages use pathological score in Duodenum and only jointly eligible FOVs contribute.

Metric strength compared with rule prevalence

Prevalence and Lift expose different changes: prevalence asks how often a rule is found among eligible FOVs, whereas Lift measures its strength only where it is found. The screens below first rank every rule by its largest observed stage gap; rule names are not chosen in advance. For avoidance, a smaller Lift means stronger spatial separation.

Colon SMV → Endothelial is the clearest attractive-strength result: pathological median Lift falls from about 3.4 in Control to 2.2 in Severe disease (ordered Lift FDR 0.015), while attractive prevalence changes much less. The method therefore detects weakening of a persisting spatial relationship, not merely disappearance of the rule.

Evidence strength: Strong.

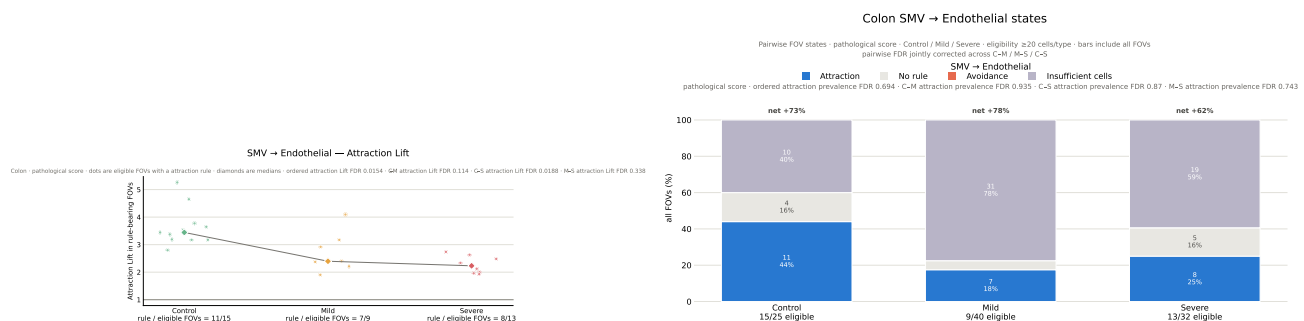


Figure 33: In the left plot, each dot is attraction Lift in one eligible rule-bearing FOV and each diamond is the stage median; the x-axis reports rule-bearing / eligible FOVs. In the right plot, the bars classify every FOV, while the printed tests compare the percentage of eligible FOVs containing an attractive rule. Stages use pathological score in Colon. Lift FDR and attraction-prevalence FDR are separate corrected tests; see the [test guide above](#).

The same rule is again the largest attractive-Lift gap when Colon biopsies are regrouped by clinical score. Its corrected evidence is weaker (best Lift FDR 0.065), but the independent grouping preserves the loss of strength.

Evidence strength: Moderate.

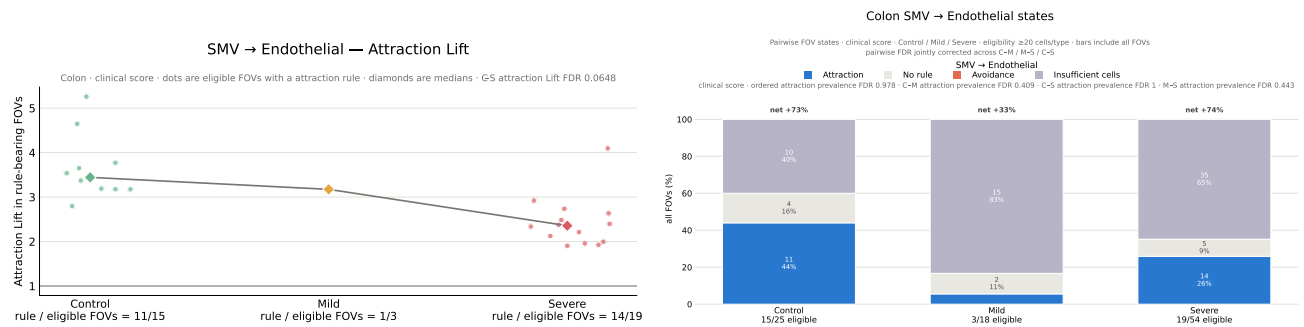


Figure 34: Values follow Figure 33; stages now use clinical score in Colon.

Representative pathological fields make the attractive SMV–Endothelial relation directly inspectable without selecting visually extreme FOVs.

Colon SMV → Endothelial — representative FOVs

pathological score · median-strength rule-bearing FOV in each stage · eligibility ≥20 cells/type

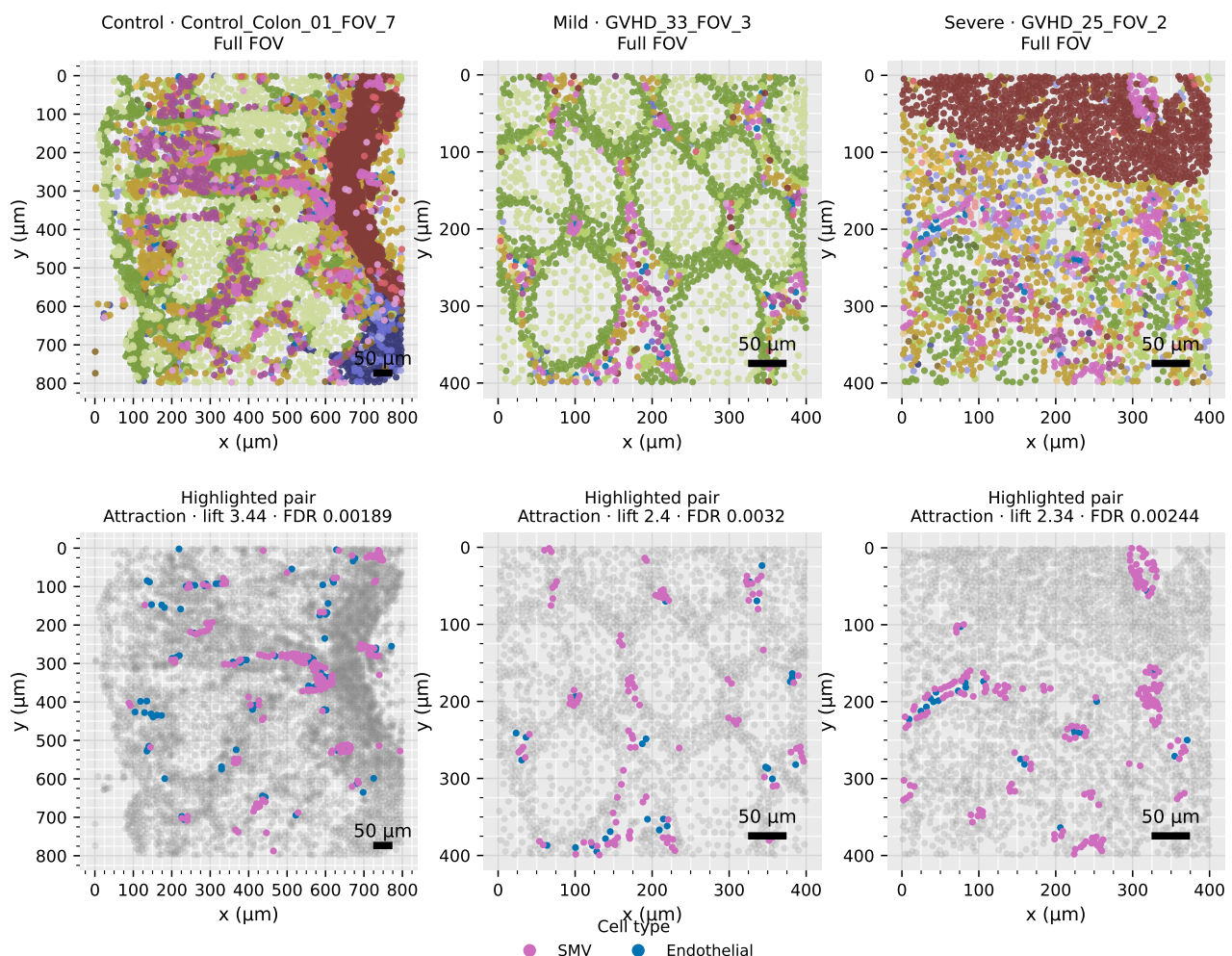


Figure 35: Within each stage, only attractive occurrences are considered and the FOV whose Lift is closest to that stage's median is shown. Row meanings follow Figure 2.

Colon Plasma → Goblet shows the converse pattern for a negative rule: avoidance becomes much stronger, with median Lift falling from about 0.73 to 0.22 (Control–Severe Lift FDR 0.008), even as avoidance prevalence changes only modestly. Thus a prevalence-only analysis would miss the strongest part of this result.

Evidence strength: Strong.

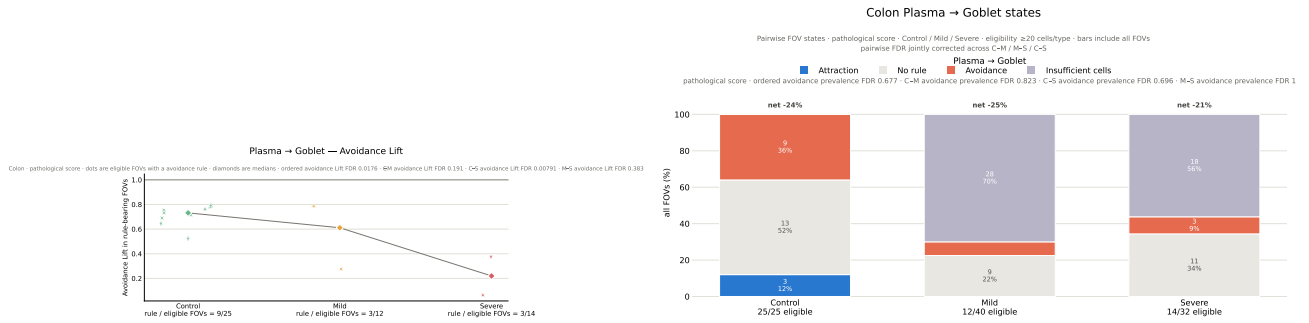


Figure 36: Values follow Figure 33, now for avoidance under pathological staging in Colon; lower Lift means stronger avoidance.

The corresponding avoidance-only representative fields show the spatial separation behind those aggregate Lift values while retaining the full tissue context.

Colon Plasma → Goblet — representative FOVs

pathological score · median-strength rule-bearing FOV in each stage · eligibility ≥20 cells/type

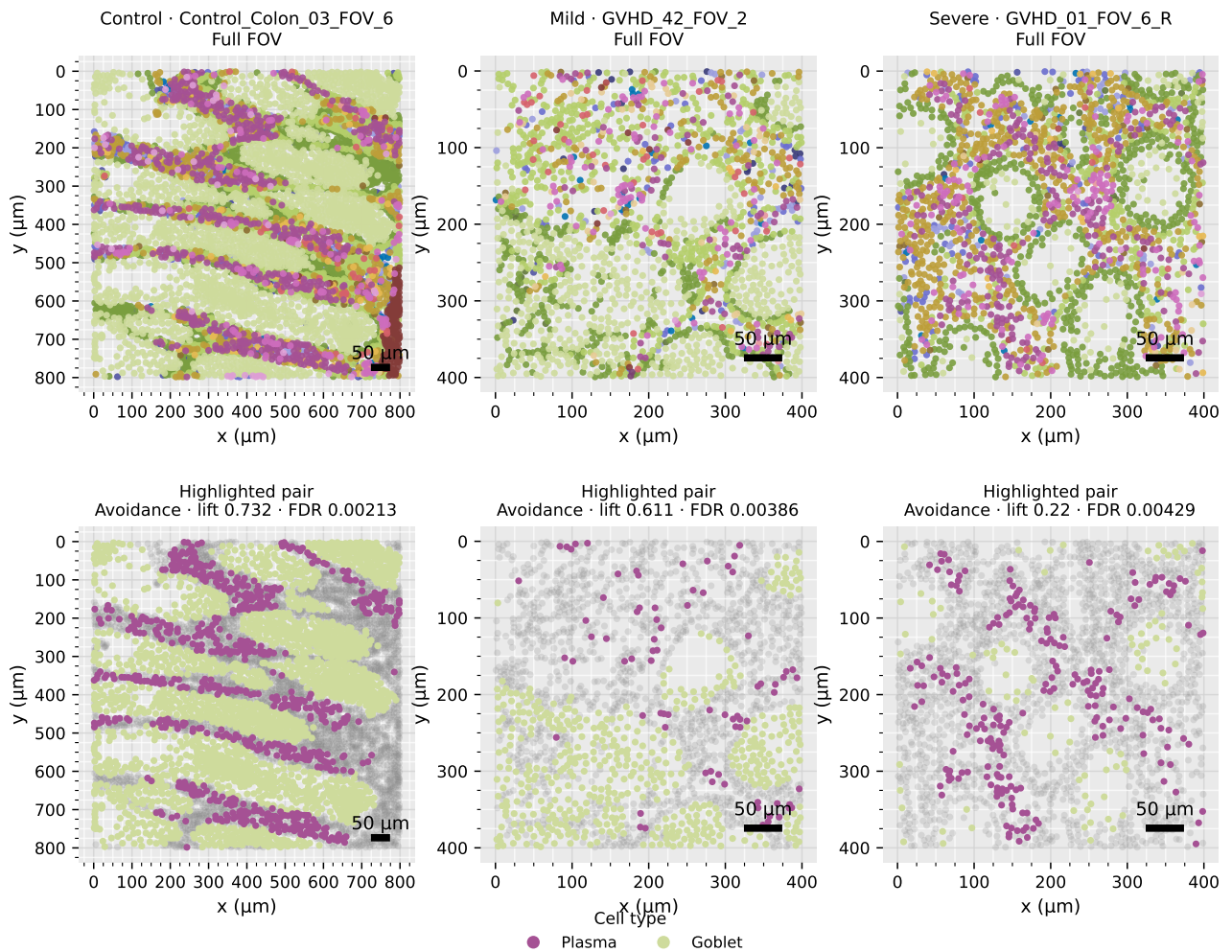


Figure 37: State-specific selection follows Figure 35; only avoidance occurrences are eligible for display.

Duodenal Muscle → CD4T changes in both dimensions: avoidance prevalence rises from 22% in Control to 62% in Mild and 58% in Severe, while median Lift falls from about 0.73 to 0.53–0.58. The largest corrected Lift comparison is borderline (FDR 0.095), so this is a coherent but exploratory immune–muscle lead.

Evidence strength: Moderate; borderline corrected evidence.

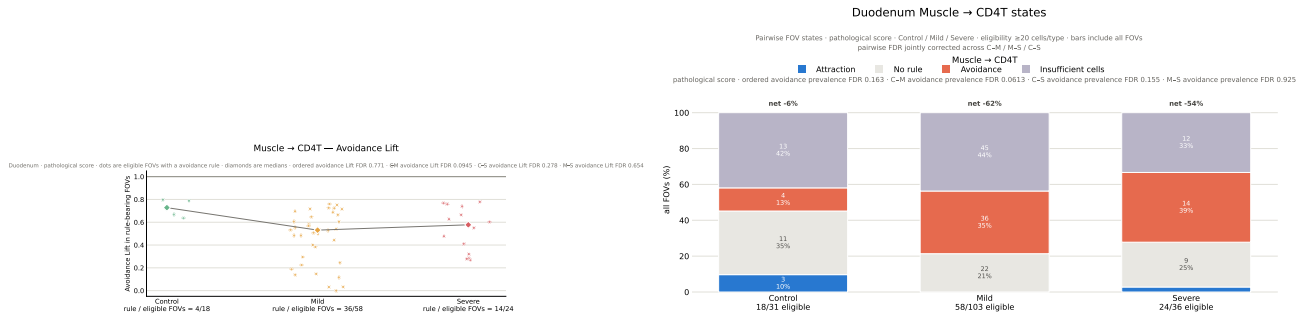


Figure 38: Values follow Figure 33; stages use pathological score in Duodenum and lower Lift means stronger avoidance.

Representative avoidance fields show that the numerical change corresponds to visible Muscle-CD4T spatial examples, but they do not establish a cell-interaction mechanism.

Duodenum Muscle → CD4T — representative FOVs

pathological score · median-strength rule-bearing FOV in each stage · eligibility ≥ 20 cells/type

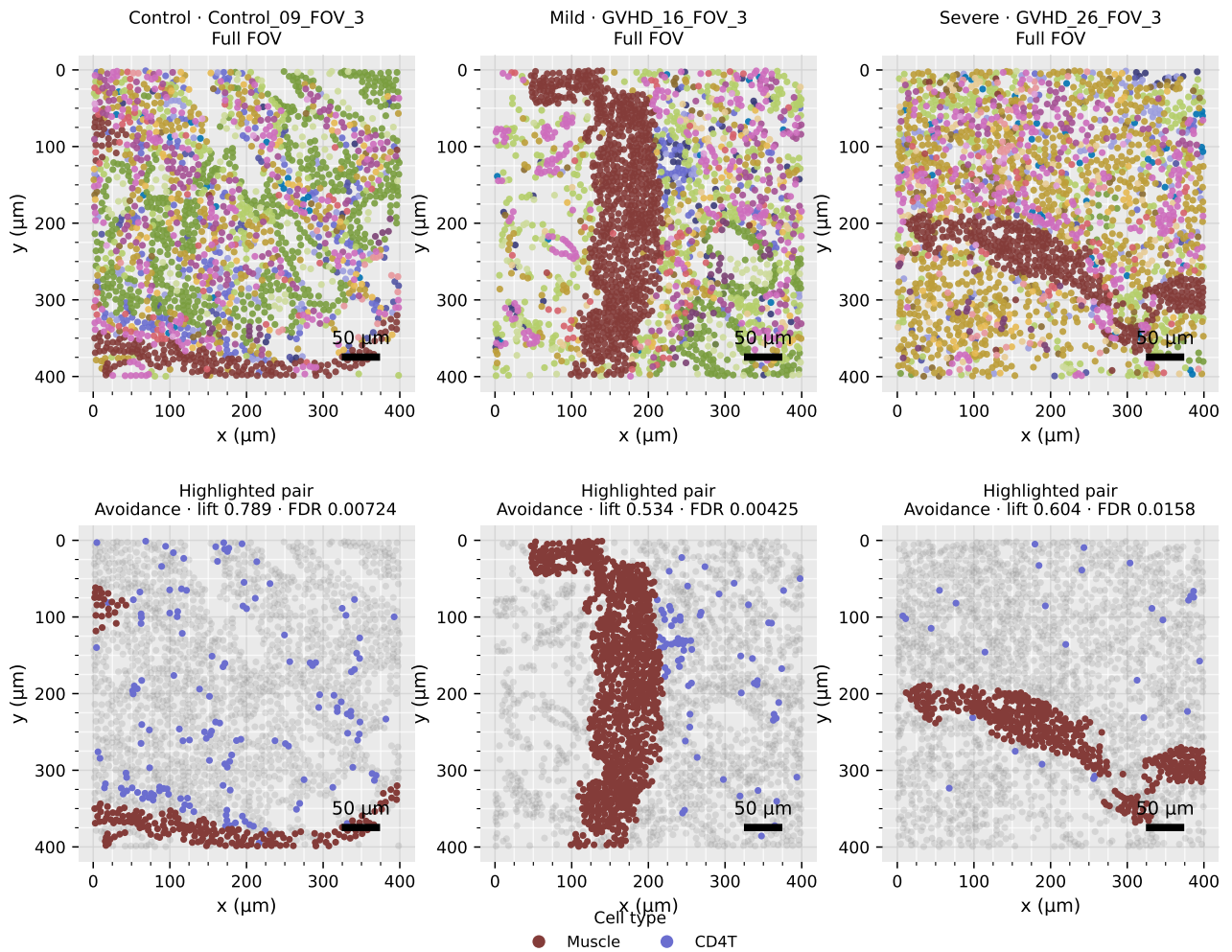


Figure 39: State-specific selection follows Figure 35.

Duodenal Goblet → Mesenchymal_VIM is selected by avoidance prevalence rather than Lift: prevalence rises from 0/3 eligible Control FOVs to 19/46 Mild and 3/4 Severe FOVs (ordered prevalence FDR 0.014), but avoidance Lift does not strengthen in parallel. This is an interesting epithelial-mesenchymal lead whose tiny endpoint denominators remain visible.

Evidence strength: Exploratory; strong trend test but sparse endpoints.

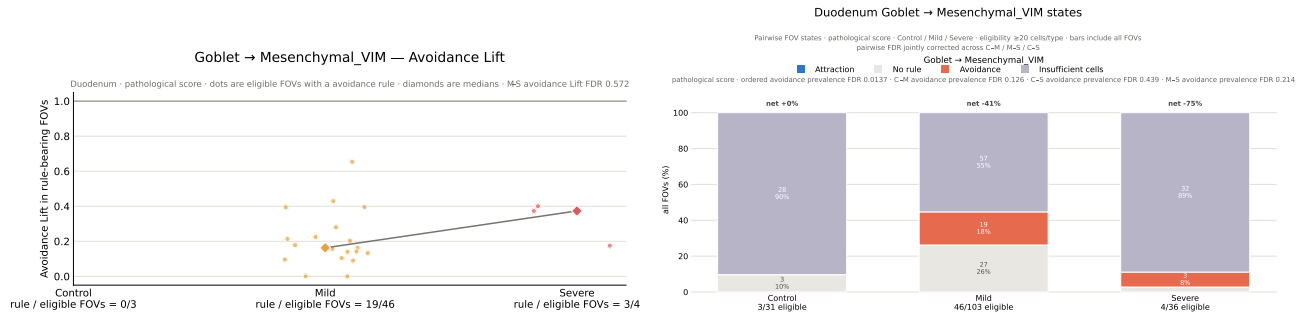


Figure 40: Values follow Figure 33; selection was based on the largest avoidance-prevalence gap, not on the named cell pair or its Lift.

The state-specific fields show why this result must remain qualified: Control has no avoidance occurrence to display and the Severe estimate rests on three avoidance FOVs.

Duodenum Goblet → Mesenchymal_VIM — representative FOVs

pathological score · median-strength rule-bearing FOV in each stage · eligibility ≥20 cells/type

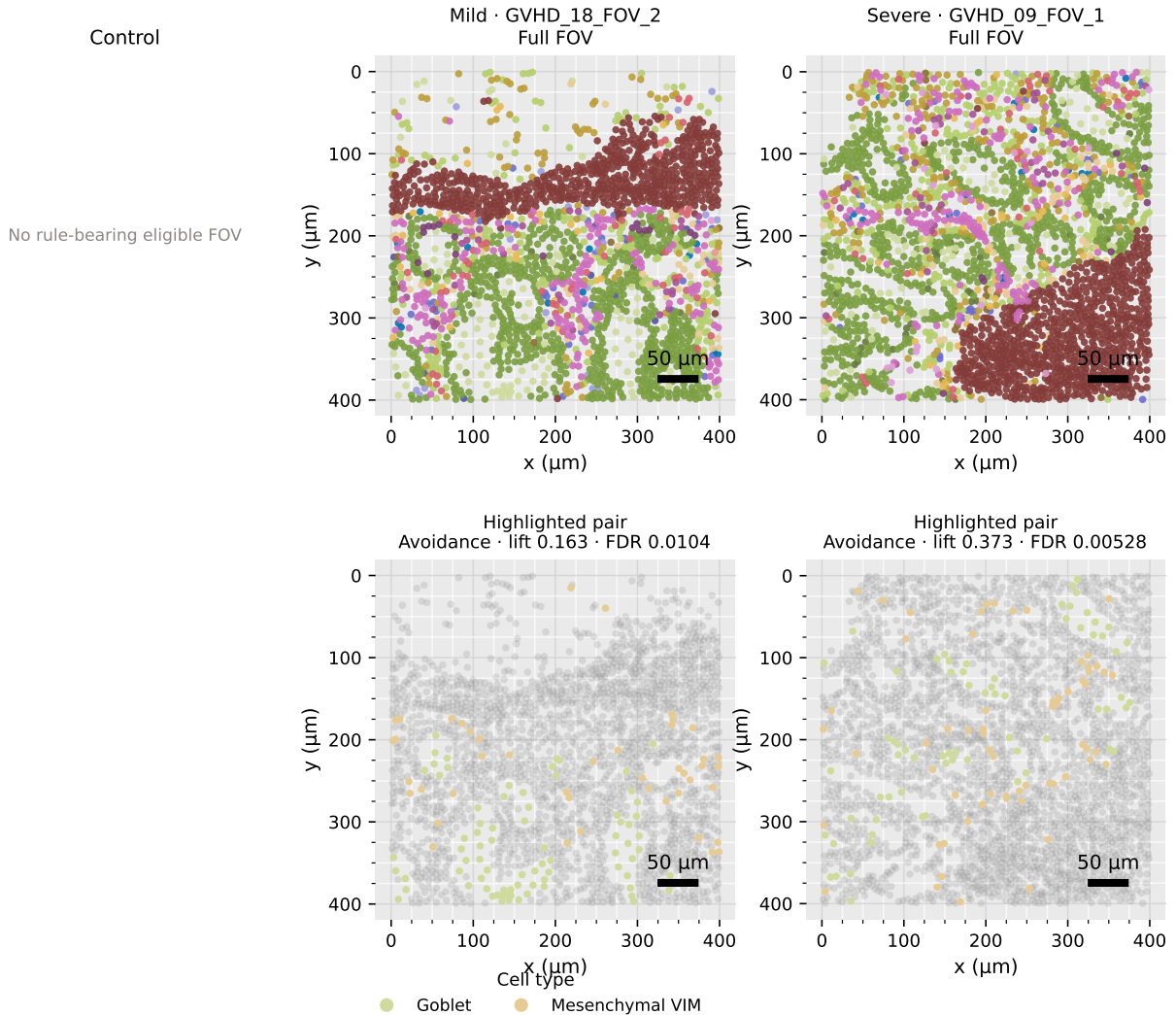


Figure 41: State-specific selection follows Figure 35; a missing stage means no eligible FOV contained the requested avoidance rule.

The largest Colon pathological avoidance-prevalence gap is Goblet → CD3T: avoidance is found in 8/12 Control and 5/5 Mild eligible FOVs but 0/6 Severe FOVs (Mild–Severe FDR 0.044). Its size is striking, yet the small denominators and non-monotonic trajectory make it a follow-up lead rather than a stable severity rule.

Evidence strength: Exploratory; corrected pairwise signal with small groups.

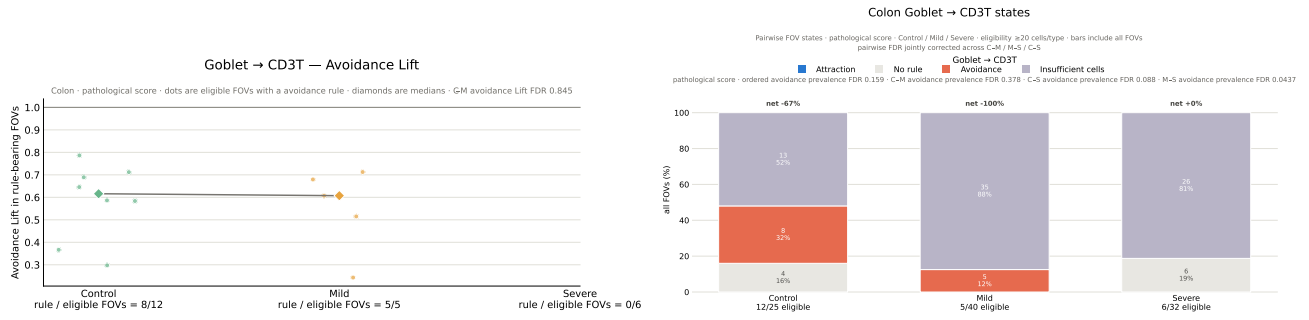


Figure 42: Values follow Figure 33; the Severe metric panel is empty because no eligible Severe FOV contains this avoidance rule.

The FOV sequence makes the same limitation concrete: avoidance examples exist in Control and Mild, but no representative Severe avoidance field can be selected.

Colon Goblet → CD3T — representative FOVs

pathological score · median-strength rule-bearing FOV in each stage · eligibility ≥20 cells/type

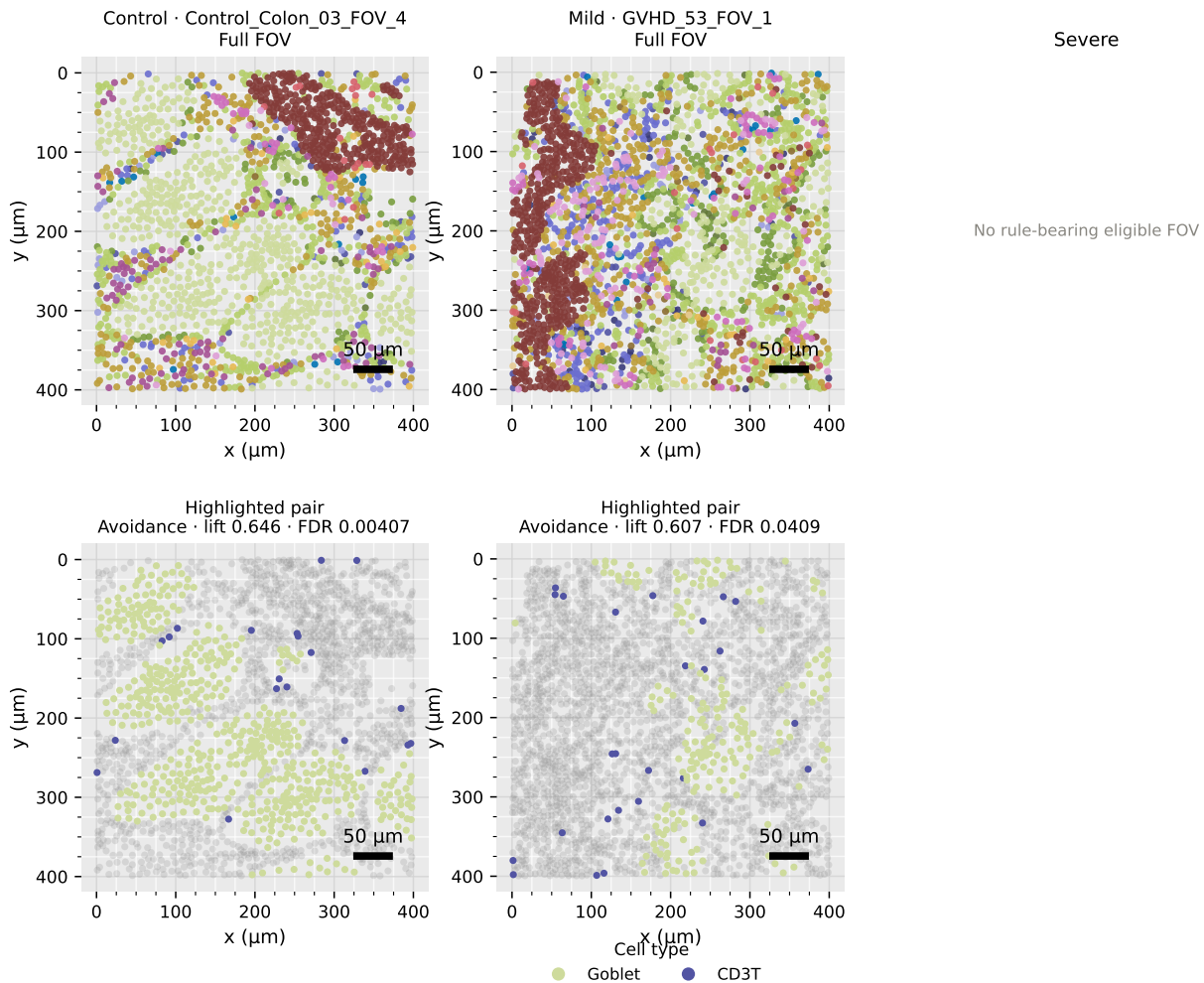


Figure 43: State-specific selection follows Figure 35.

Under clinical staging, Colon Treg → Goblet has the largest avoidance-prevalence gap (1/7, 4/4 and 0/9 eligible FOVs), but the pattern is non-monotonic and driven by only four Mild FOVs. It is retained because it is an immune–epithelial lead, not because it is robust.

Evidence strength: Weak but interesting; Mild–Severe FDR 0.044.

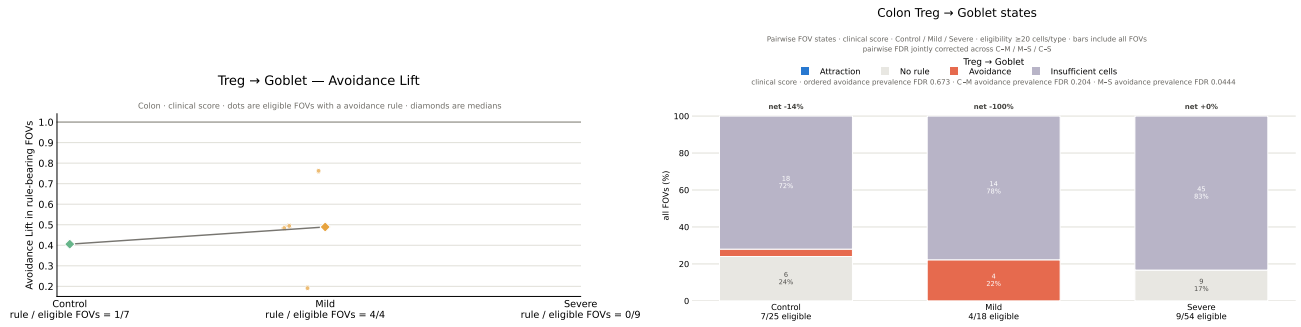


Figure 44: Values follow Figure 33; selection was by the largest clinical avoidance-prevalence gap in Colon.

Representative fields exist only for Control and Mild, visibly separating the observed examples from the absent Severe avoidance state.

Colon Treg → Goblet — representative FOVs

clinical score · median-strength rule-bearing FOV in each stage · eligibility ≥20 cells/type

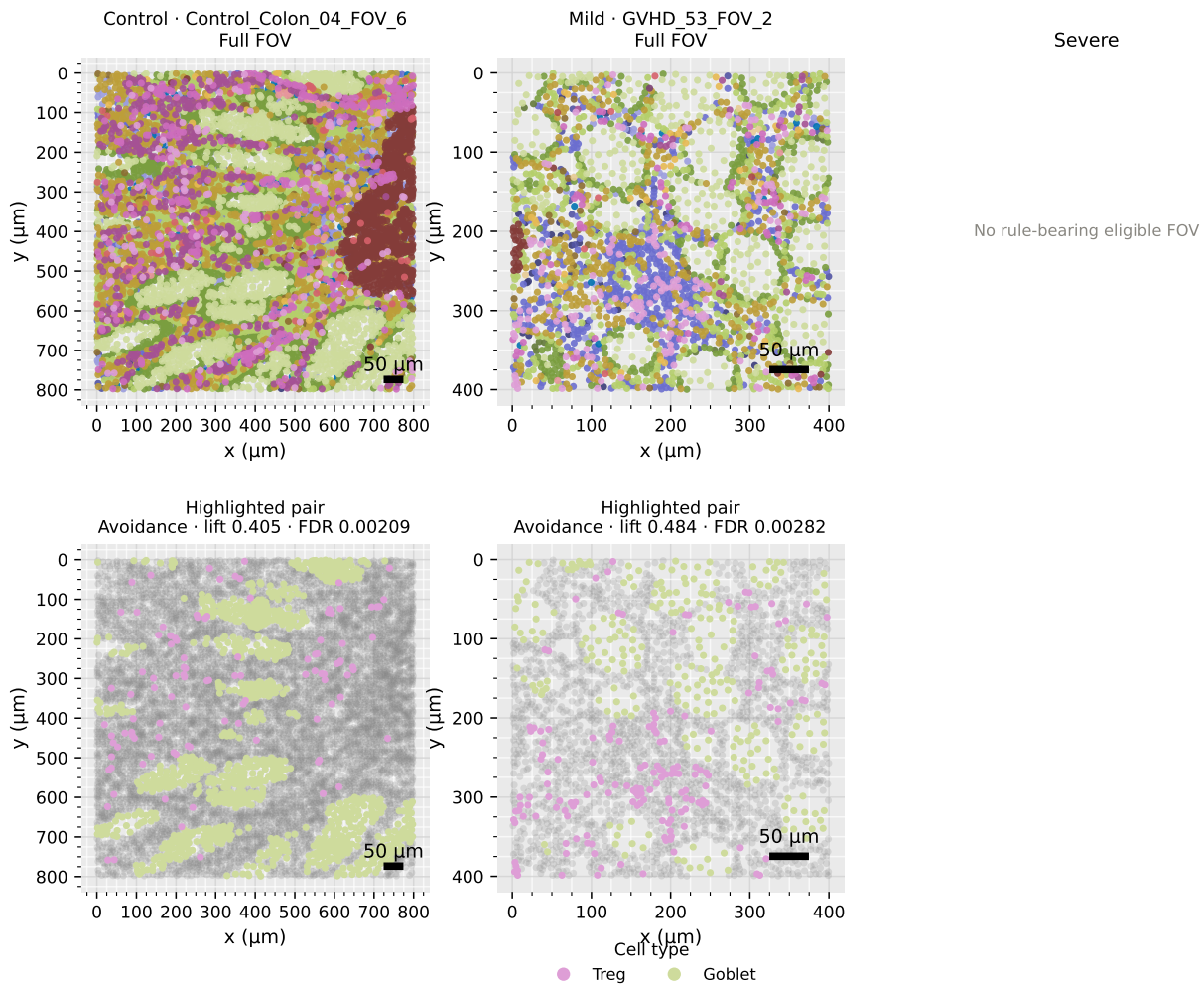


Figure 45: State-specific selection follows Figure 35.

Exploratory changes after transplantation

Although it does not pass correction (temporal FDR 0.335), Colon CD8T → Goblet shifts from a mildly avoidant Control context toward markedly stronger avoidance after 100 days, making it the clearest immune–epithelial temporal lead in Colon.

Evidence strength: Exploratory; not statistically significant.

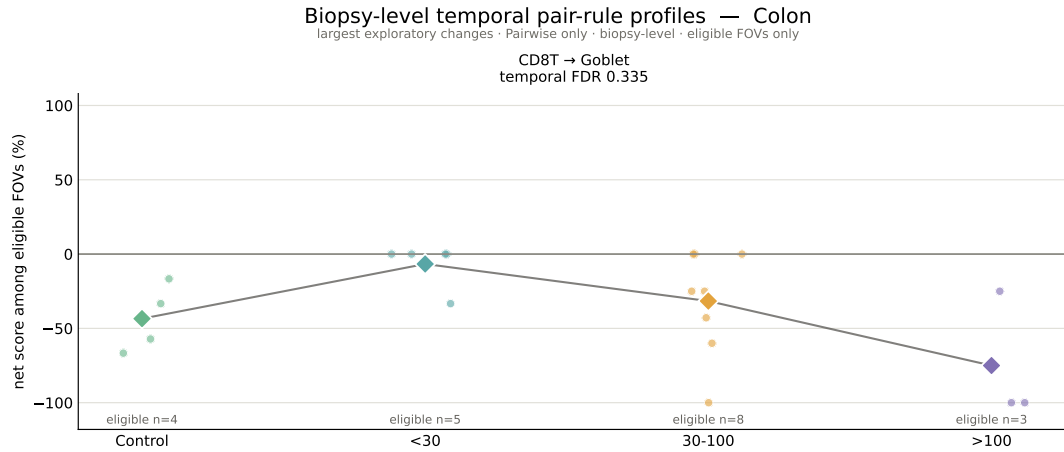


Figure 46: Each dot is one biopsy's mean rule state across eligible FOVs, multiplied by 100; diamonds are time-group means and eligible n is the contributing biopsy count. Temporal FDR tests the largest difference among < 30, 30–100 and > 100 days and is corrected across all tested Colon pair rules; Control is not part of that test. See the [test guide above](#).

The corresponding tissue maps show typical rule-bearing biopsies across the same windows; they make the CD8T–Goblet context visible, but remain illustrative and do not change the non-significant temporal FDR of 0.335.

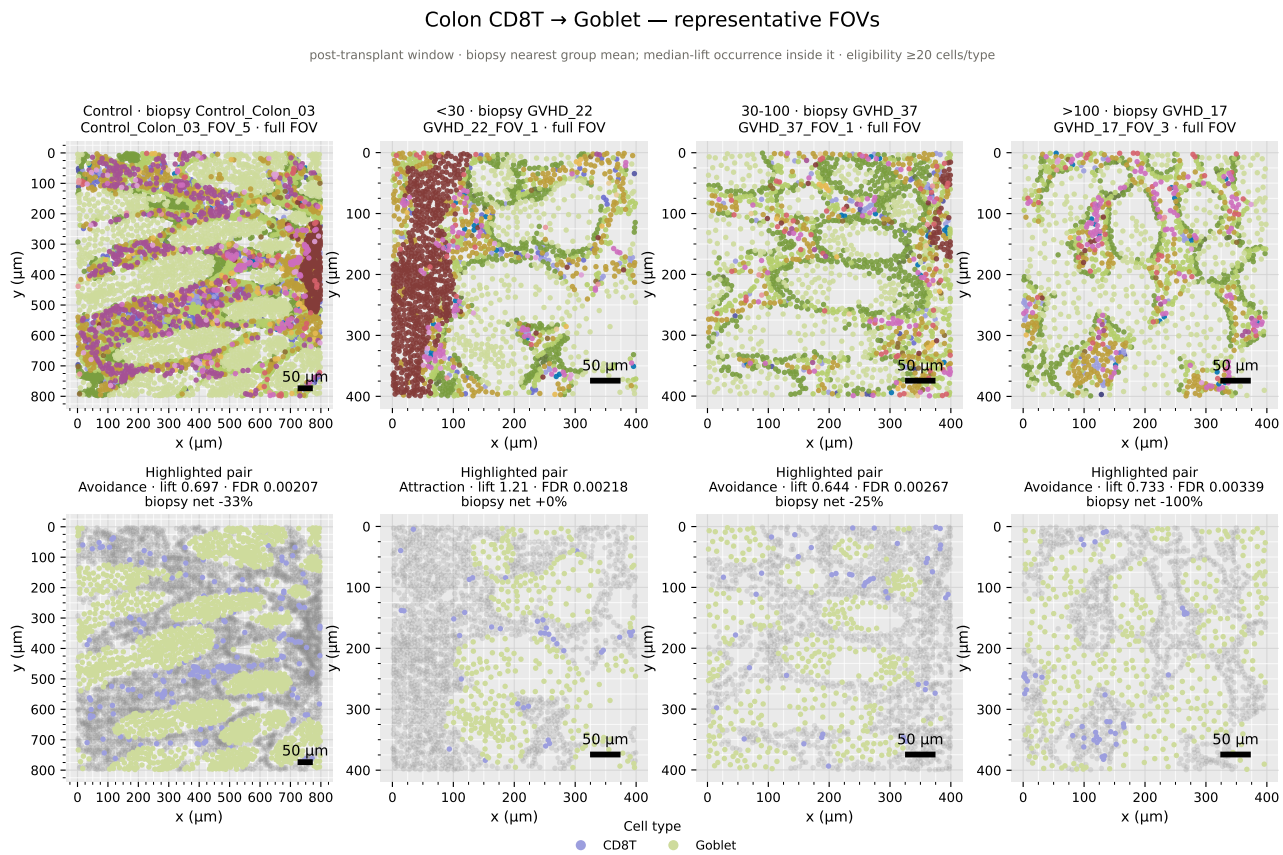


Figure 47: For each displayed group, selection first finds the rule-bearing biopsy whose biopsy net score is closest to the mean across all eligible biopsies in that group. Inside it, the more frequent nonzero rule direction is retained and the occurrence nearest its median Lift is shown. The upper row contains every cell and the lower row highlights the named pair. Control is contextual and is not included in the temporal test.

Colon SMV → Endothelial loses much of its attractive net score after 100 days; the trajectory is large and vascularly relevant, but remains exploratory because temporal FDR is 0.362.

Evidence strength: Exploratory; not statistically significant.

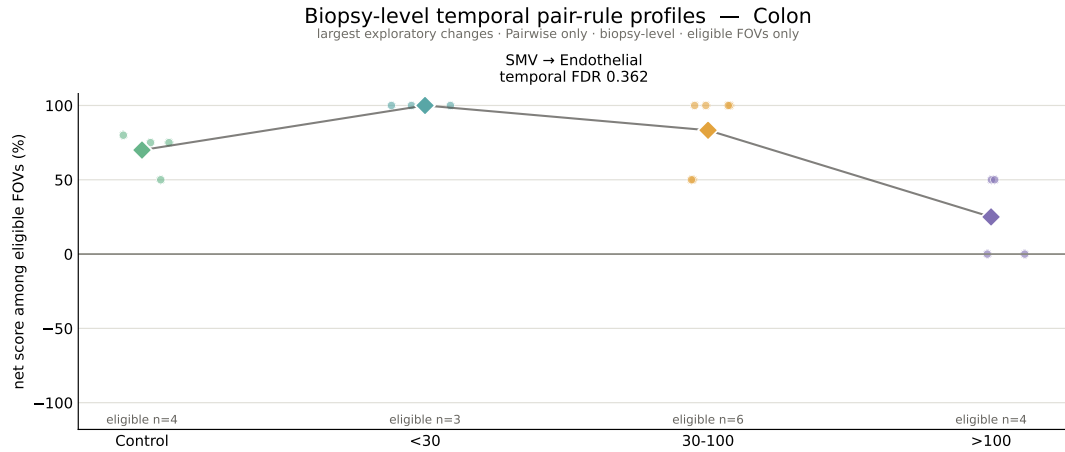


Figure 48: Values follow Figure 46; this is a separately tested directional Colon rule.

The representative biopsy/FOV sequence shows the vascular cell pair in its tissue setting; it is useful spatial context, but it does not make the temporal FDR of 0.362 significant.

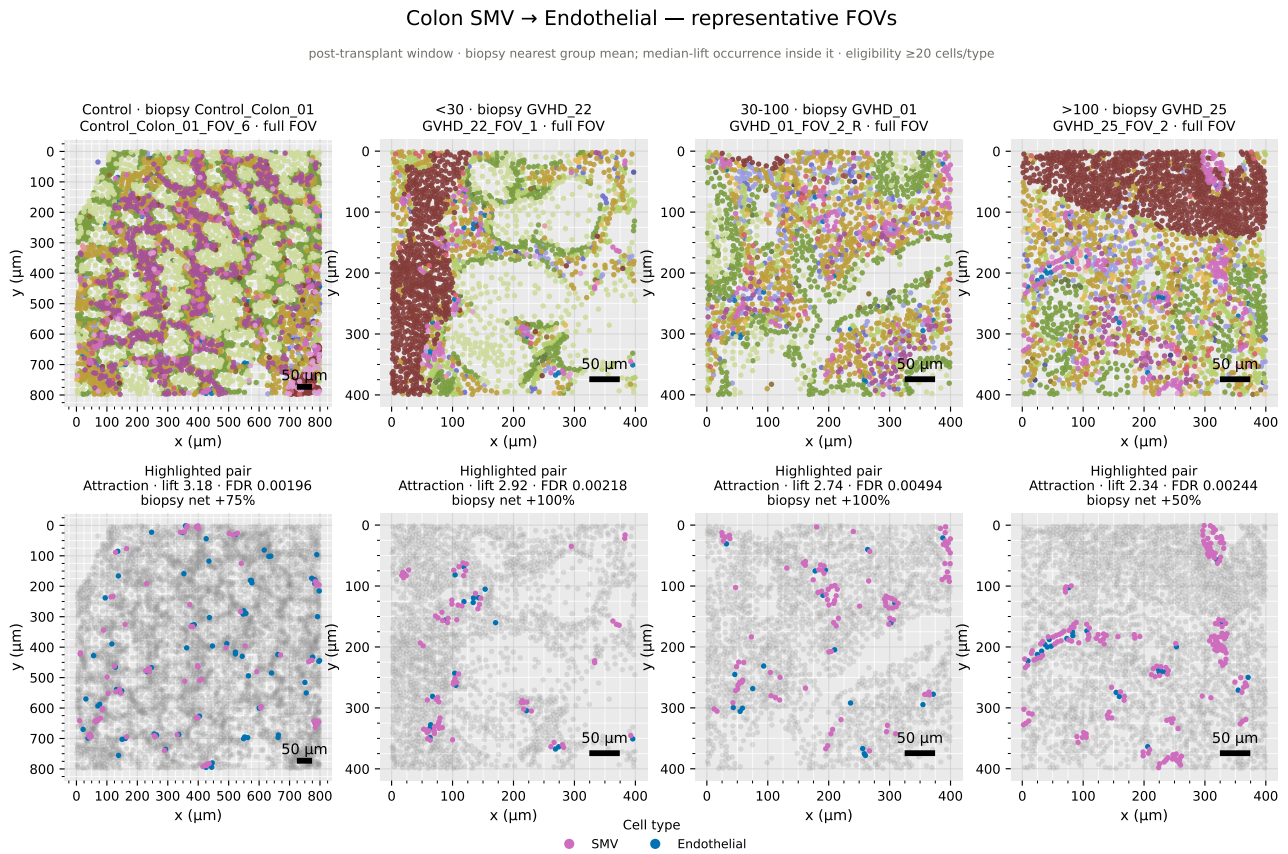


Figure 49: Selection and row meanings follow Figure 47; the individual FOV's state, Lift and mining FDR are printed above each highlighted map.

Colon SMV → Goblet becomes more avoidant again after an early relaxation, suggesting a possible late epithelial-vascular change; temporal FDR is 0.362, so this is a follow-up lead rather than confirmed evidence.

Evidence strength: Exploratory; not statistically significant.

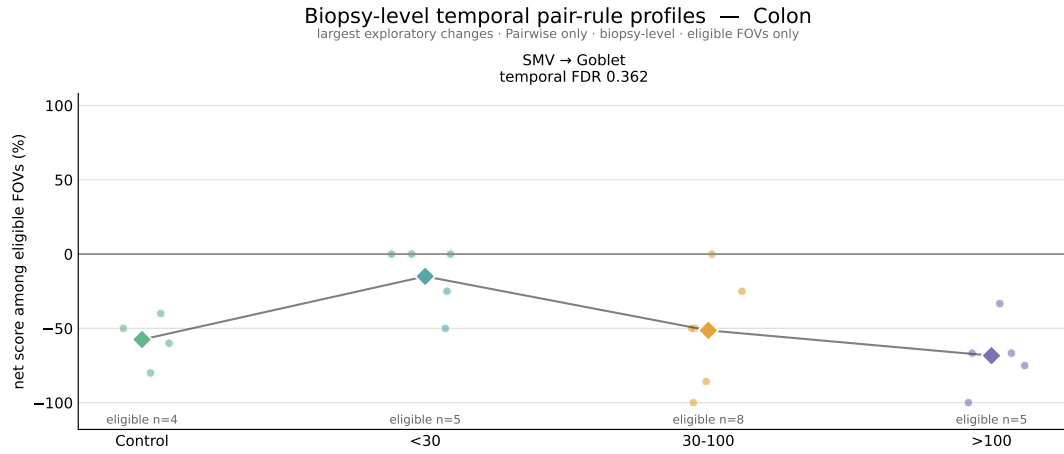


Figure 50: Values follow Figure 46; the same biopsy-level calculation and eligibility threshold are used.

These automatically selected maps show where the SMV–Goblet examples occur within typical rule-bearing biopsies; they illustrate the proposed late change but do not overcome temporal FDR 0.362.

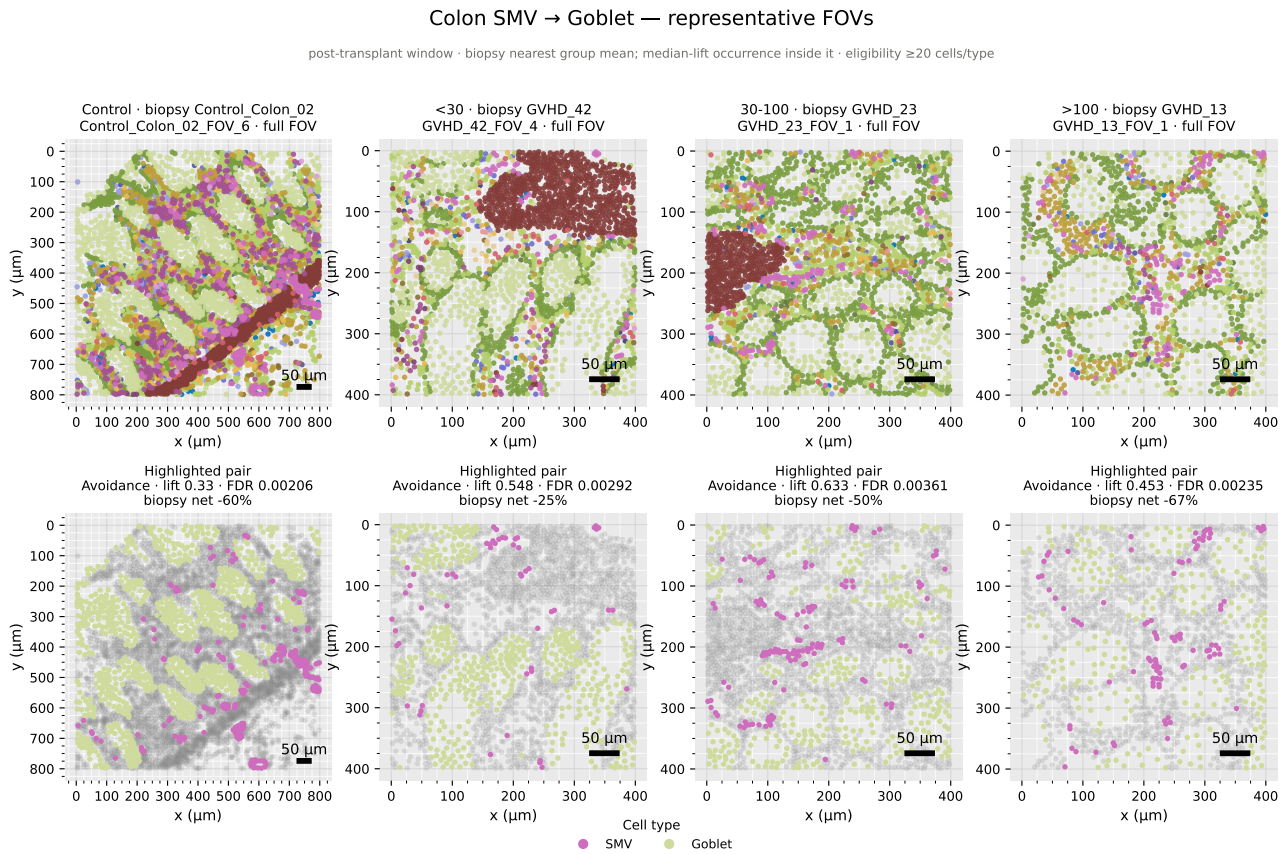


Figure 51: Selection and row meanings follow Figure 47.

In Duodenum, Plasma_CD38 → Fibroblast is the closest pooled temporal lead: the < 30 versus ≥ 30 -day comparison has a 34-percentage-point gap and joint FDR 0.086, which is suggestive but does not pass 0.05.

Evidence strength: Exploratory; near-significant but not statistically significant.

Pooled temporal contrasts — Duodenum

Pairwise only · biopsy-level · eligible FOVs only · joint FDR: 2 contrasts, 431 tests

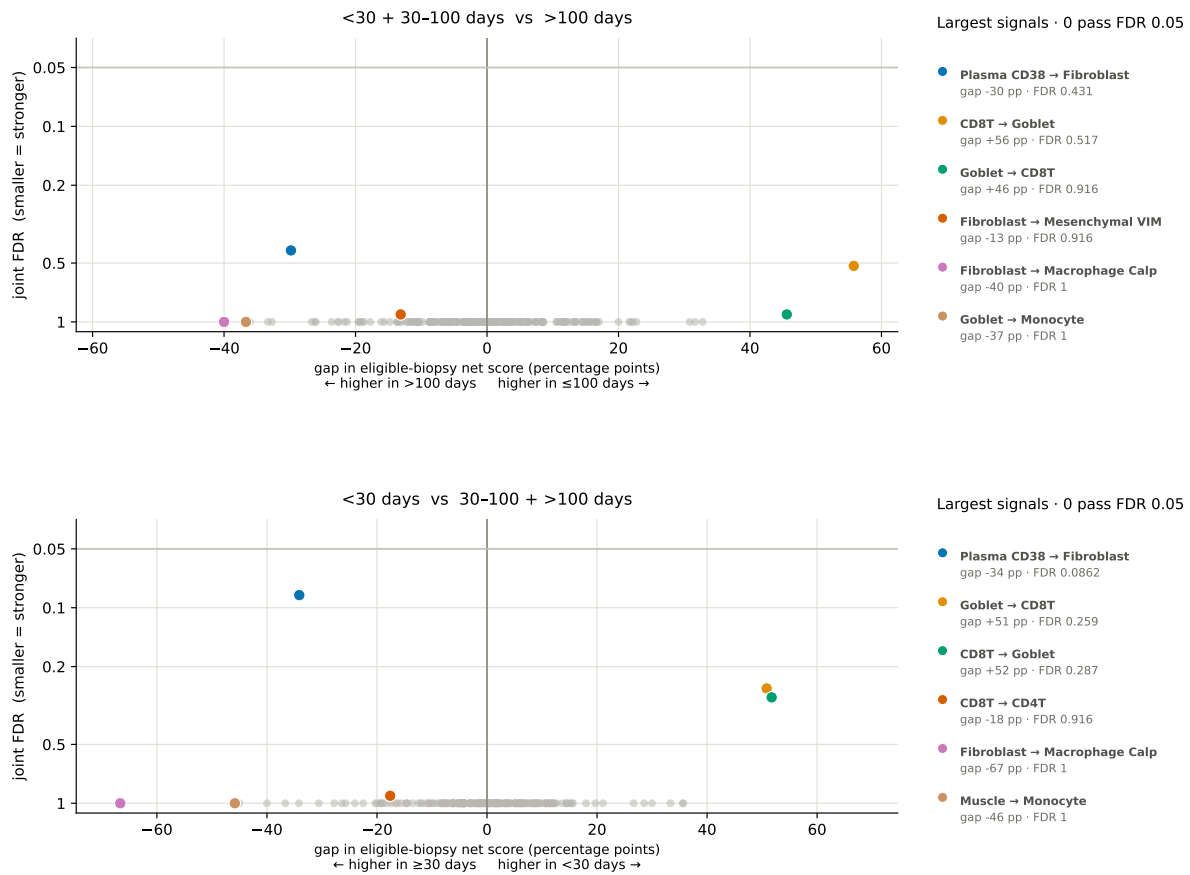


Figure 52: Each point is one directional pair rule. The x-axis is the difference between the two named groups in biopsy-level net score, in percentage points; the y-axis is joint FDR. The lower panel contains the Plasma_CD38 → Fibroblast result. FDR is corrected jointly across both pooled contrasts and every tested Duodenal pair rule.

The stage-resolved FOV maps provide a concrete view of the Plasma_CD38–Fibroblast fields behind this pooled lead; they remain illustrative because joint FDR 0.086 does not pass 0.05.

Duodenum Plasma_CD38 → Fibroblast — representative FOVs

post-transplant window · biopsy nearest group mean; median-lift occurrence inside it · eligibility ≥ 20 cells/type

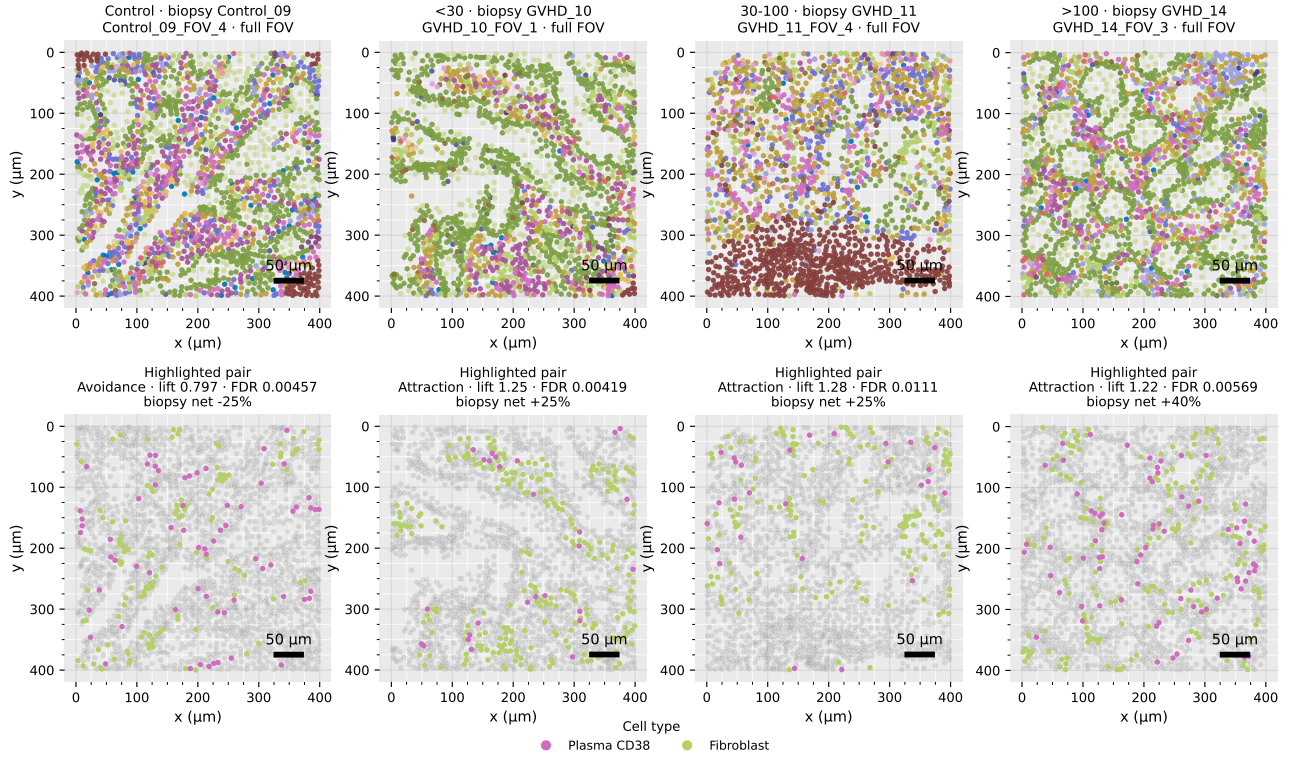


Figure 53: Selection and row meanings follow Figure 47; the maps retain the original four display groups even though the statistical lead pools 30–100 and > 100 days into the ≥ 30 -day side of the comparison.